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Sivanesan et al.

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(54) **ADAPTIVE AND HIERARCHICAL NETWORK AUTHENTICATION FRAMEWORK**

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(51) **Int. Cl.**
H04W 12/06 (2021.01)
H04W 12/79 (2021.01)

(52) **U.S. Cl.**
CPC ***H04W 12/06*** (2013.01); ***H04W 12/79*** (2021.01)

(58) **Field of Classification Search**
CPC .. H04L 1/0025; H04W 28/04; H04W 56/001; H04W 12/06; H04W 12/79
USPC 370/336
See application file for complete search history.

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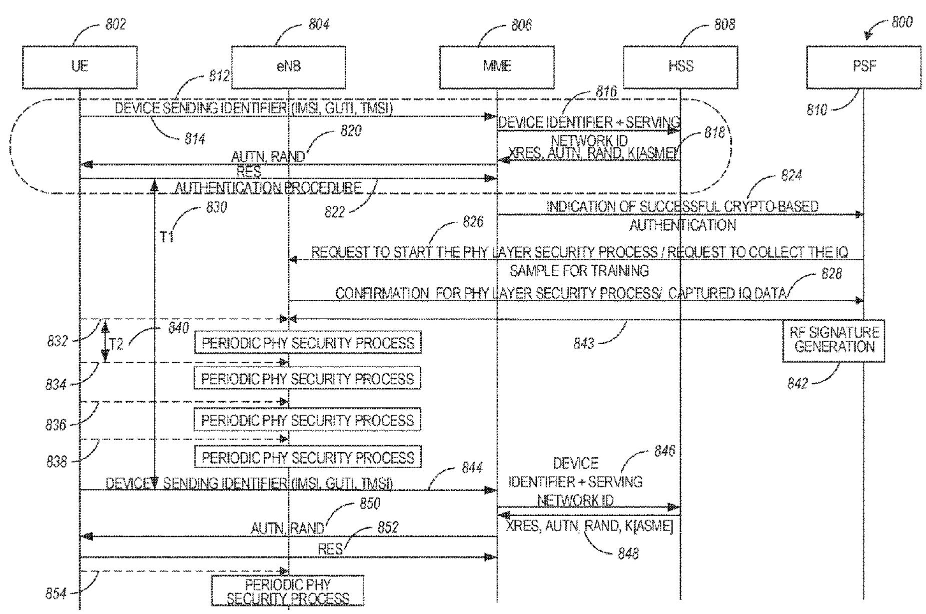
(74) *Attorney, Agent, or Firm* — Schwegman Lundberg & Woessner, P.A.

(57) **ABSTRACT**

A non-transitory computer-readable storage medium stores instructions to configure a base station for user equipment (UE) authentication in a wireless network and to cause the base station to perform an operation comprising decoding configuration signaling received from a PHY security function (PSF) of the wireless network. The configuration signaling includes a request for collection of a plurality of signal samples from the UE, the UE authenticated based on successful completion of a first authentication process. A response message is encoded for transmission to the PSF. The response message includes the plurality of UE signal samples. A trained machine learning model received from the PSF is decoded. The trained machine learning model associates the authenticated UE with an RF signature of the UE. The RF signature is based on the plurality of signal samples. A second authentication process of the UE is performed based on the trained model.

14 Claims, 11 Drawing Sheets



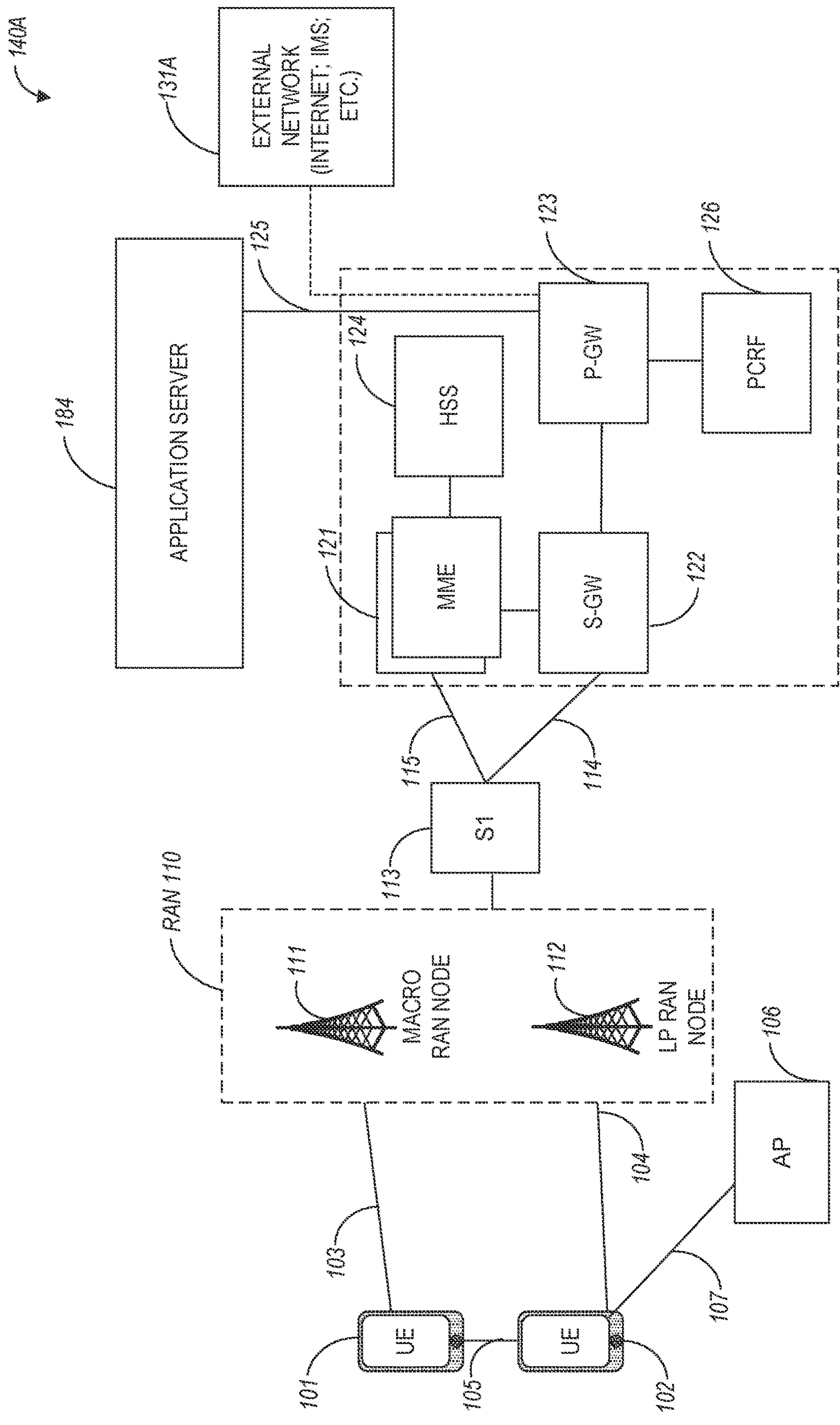

140A
EXTERNAL NETWORK (INTERNET, IMS, ETC.)
131A
APPLICATION SERVER
184
125
123
126
124
HSS
P-GW
PCRF
CN (EPC) 120
121
MME
S-GW
122
115
114
S1
113
RAN 110
111
MACRO RAN NODE
112
LP RAN NODE
106
AP
103
104
107
UE
UE
101
105
102

FIG. 1A

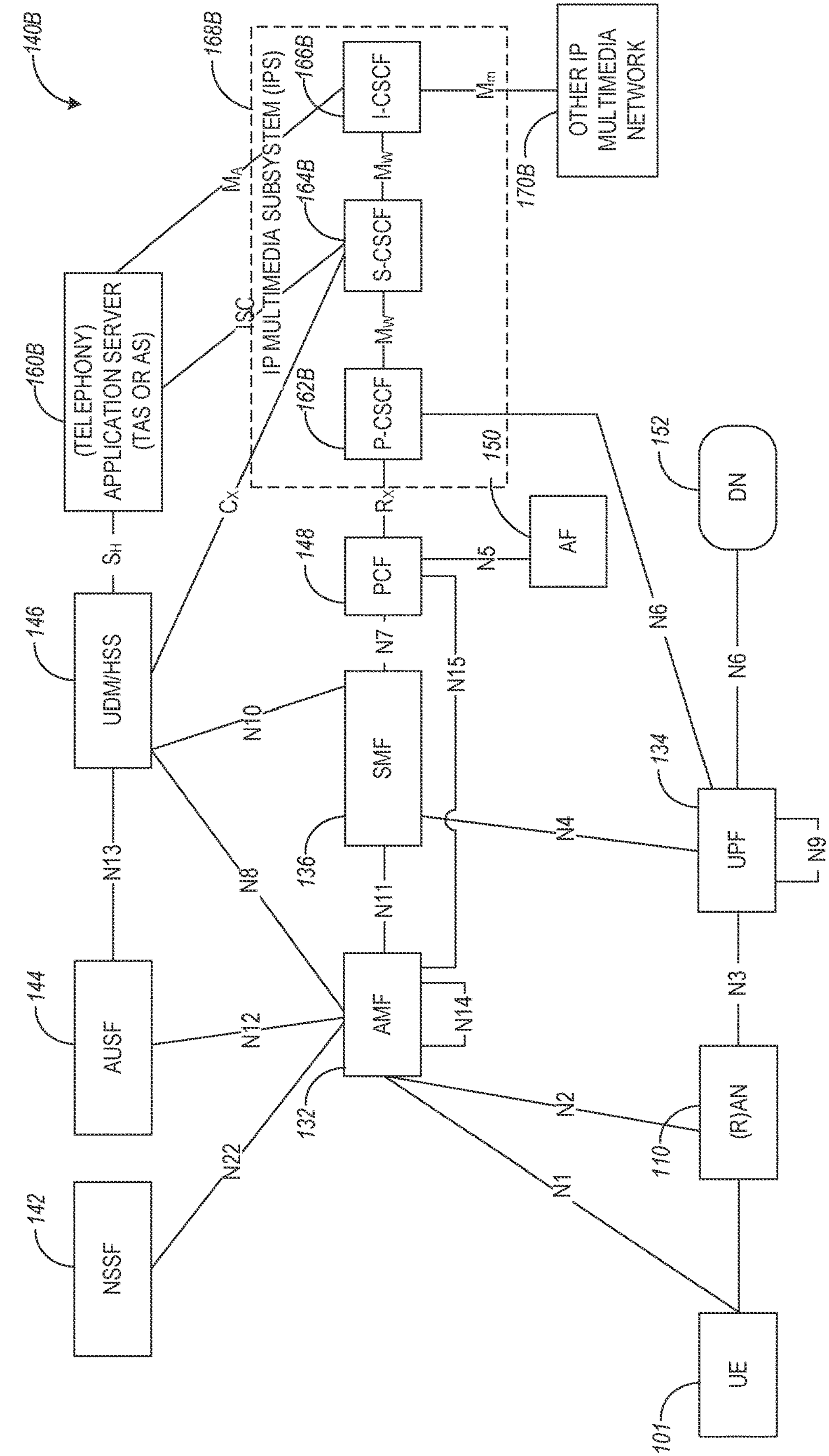

140B
168B
IP MULTIMEDIA SUBSYSTEM (IPS)
166B
I-CSCF
164B
S-CSCF
162B
P-CSCF
OTHER IP MULTIMEDIA NETWORK
170B
160B
(TELEPHONY) APPLICATION SERVER (TAS OR AS)
146
UDM/HSS
144
AUSF
142
NSSF
148
PCF
150
AF
152
DN
136
SMF
132
AMF
134
UPF
110
(R)AN
101
UE
M_A
ISC
M_W
M_m
S_H
C_X
R_X
N5
N6
N7
N15
N10
N13
N8
N11
N4
N9
N3
N12
N14
N2
N22
N1

FIG. 1B

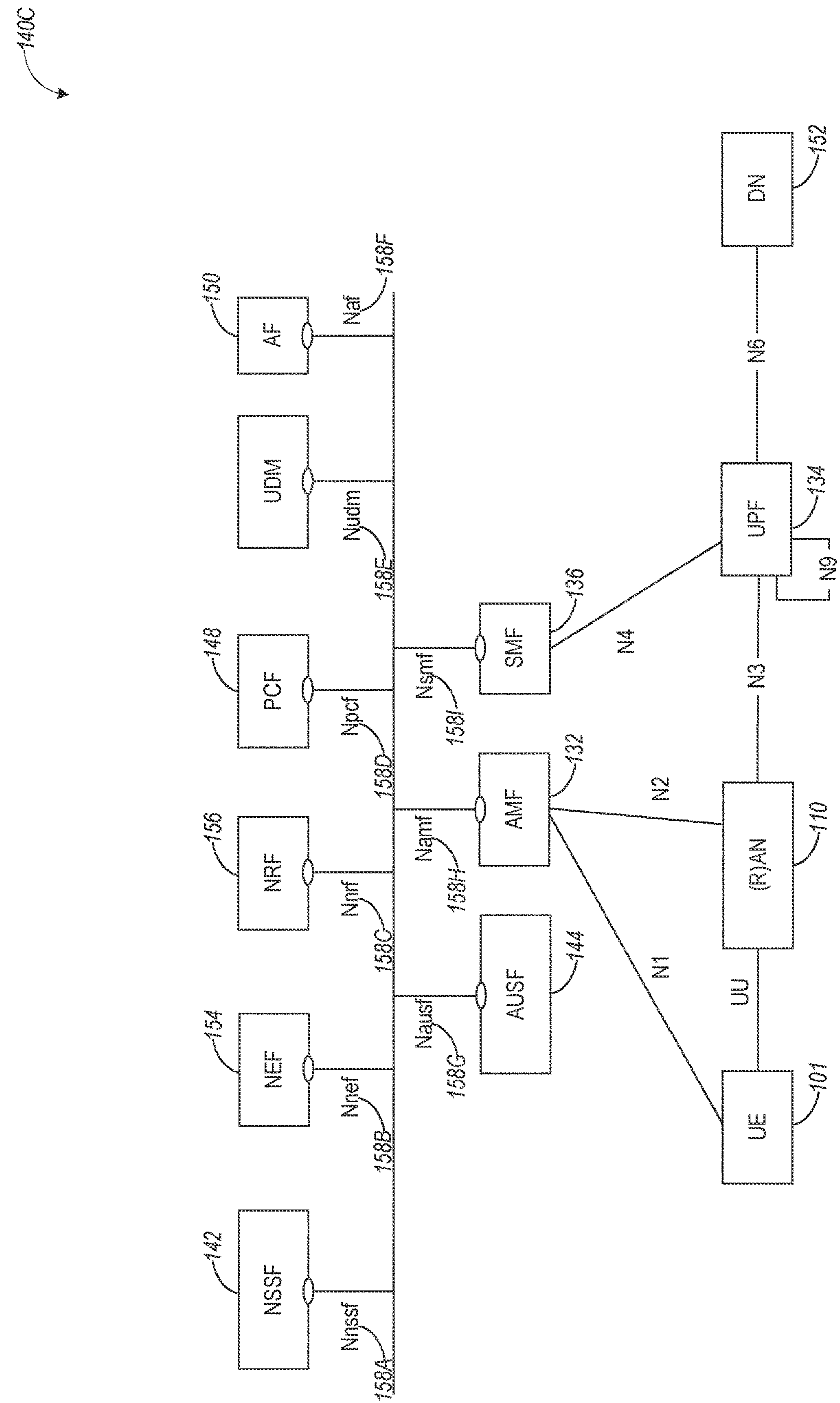
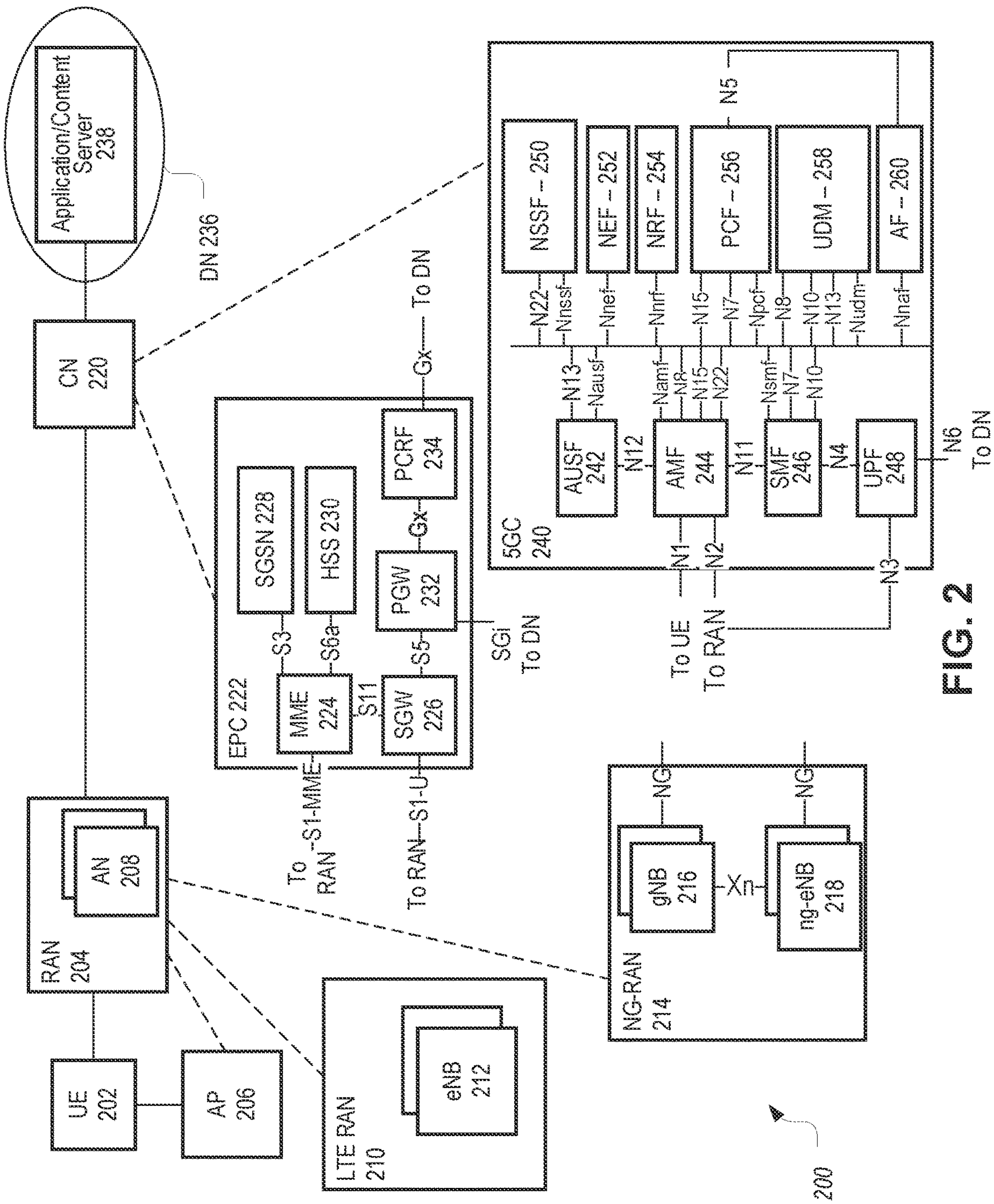

140C
NSSF
142
Nnssf
158A
NEF
154
Nnef
158B
NRF
156
Nnrf
158C
PCF
148
Npcf
158D
UDM
Nudm
158E
AF
150
Naf
158F
AUSF
144
Nausf
158G
AMF
132
Namf
158H
SMF
136
Nsmf
158I
UE
101
UU
(R)AN
110
N1
N2
N3
N4
UPF
134
N9
N6
DN
152

FIG. 1C



Application/Content Server 238
DN 236
CN 220
RAN 204
AN 208
UE 202
AP 206
LTE RAN 210
eNB 212
NG-RAN 214
gNB 216
Xn
ng-eNB 218
NG
NG
EPC 222
MME 224
SGW 226
SGSN 228
HSS 230
PGW 232
PCRF 234
S1-MME
S1-U
To RAN
To RAN
S3
S6a
S11
S5
Gx
Gx
To DN
SGi
To DN
5GC 240
AUSF 242
AMF 244
SMF 246
UPF 248
NSSF – 250
NEF – 252
NRF – 254
PCF – 256
UDM – 258
AF – 260
N22
Nnssf
Nnef
Nnrf
N15
N7
Npcf
N8
N10
N13
Nudm
Nnaf
N13
Nausf
Namf
N8
N15
N22
Nsmf
N7
N10
N12
N11
N4
N5
N6
To DN
N1
N2
N3
To UE
To RAN
200

FIG. 2

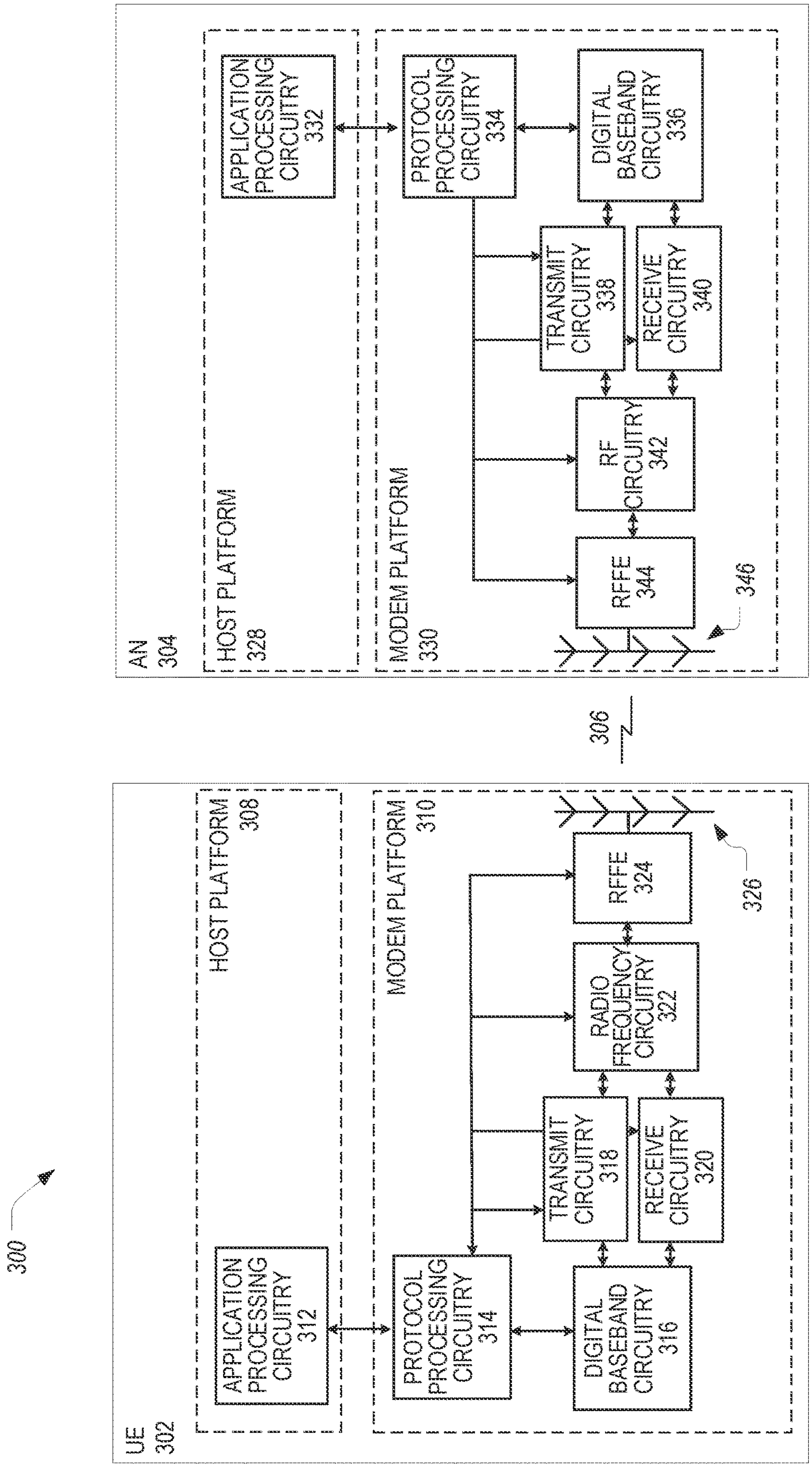
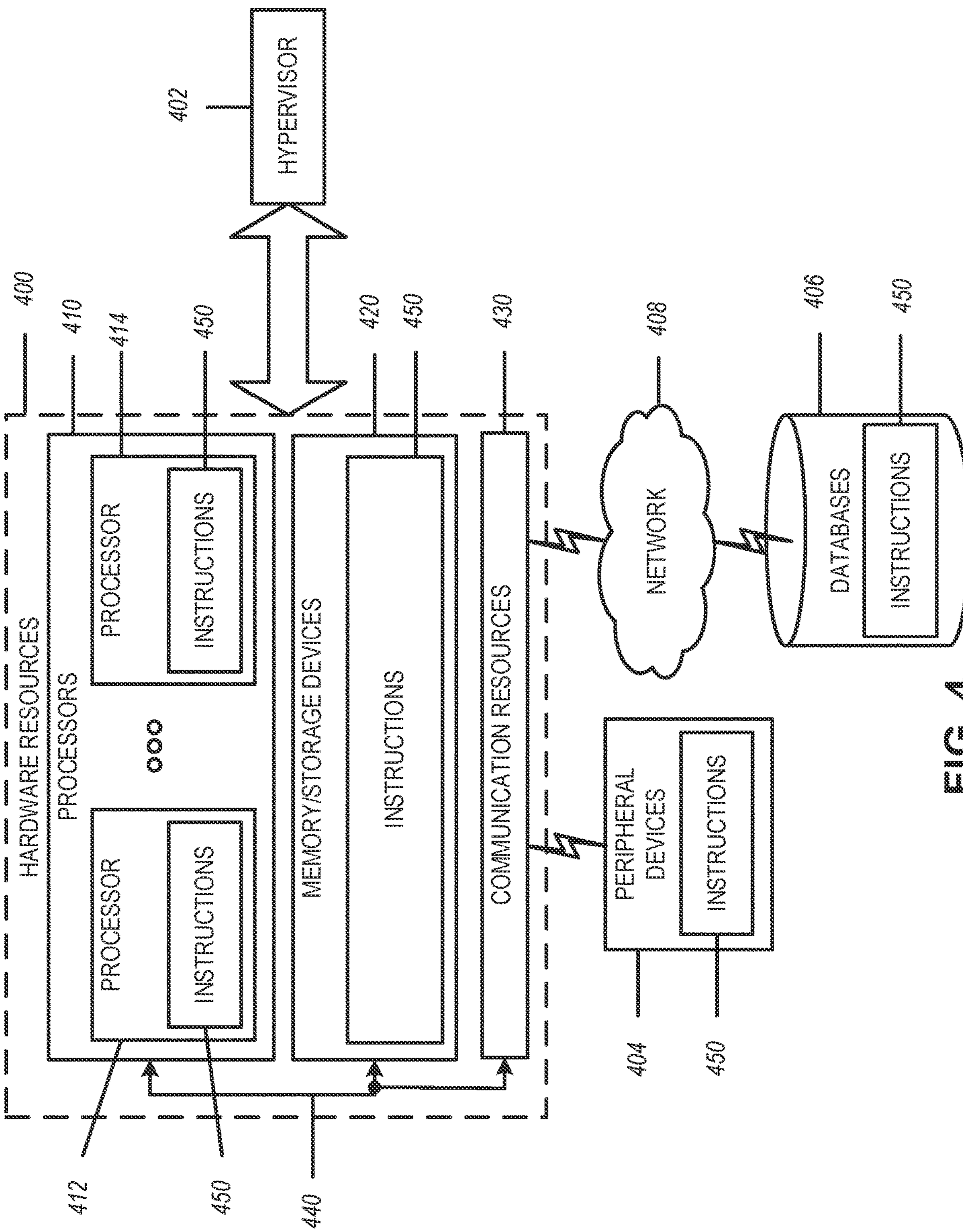

300
UE
302
HOST PLATFORM
308
APPLICATION PROCESSING CIRCUITRY
312
MODEM PLATFORM
310
PROTOCOL PROCESSING CIRCUITRY
314
DIGITAL BASEBAND CIRCUITRY
316
TRANSMIT CIRCUITRY
318
RECEIVE CIRCUITRY
320
RADIO FREQUENCY CIRCUITRY
322
RFFE
324
326
306
AN
304
HOST PLATFORM
328
APPLICATION PROCESSING CIRCUITRY
332
MODEM PLATFORM
330
PROTOCOL PROCESSING CIRCUITRY
334
DIGITAL BASEBAND CIRCUITRY
336
TRANSMIT CIRCUITRY
338
RECEIVE CIRCUITRY
340
RF CIRCUITRY
342
RFFE
344
346

FIG. 3



HYPERVISOR
402
400
410
414
450
420
450
430
HARDWARE RESOURCES
PROCESSORS
PROCESSOR
INSTRUCTIONS
PROCESSOR
INSTRUCTIONS
MEMORY/STORAGE DEVICES
INSTRUCTIONS
COMMUNICATION RESOURCES
NETWORK
408
DATABASES
INSTRUCTIONS
406
450
PERIPHERAL DEVICES
INSTRUCTIONS
404
450
412
450
440

FIG. 4

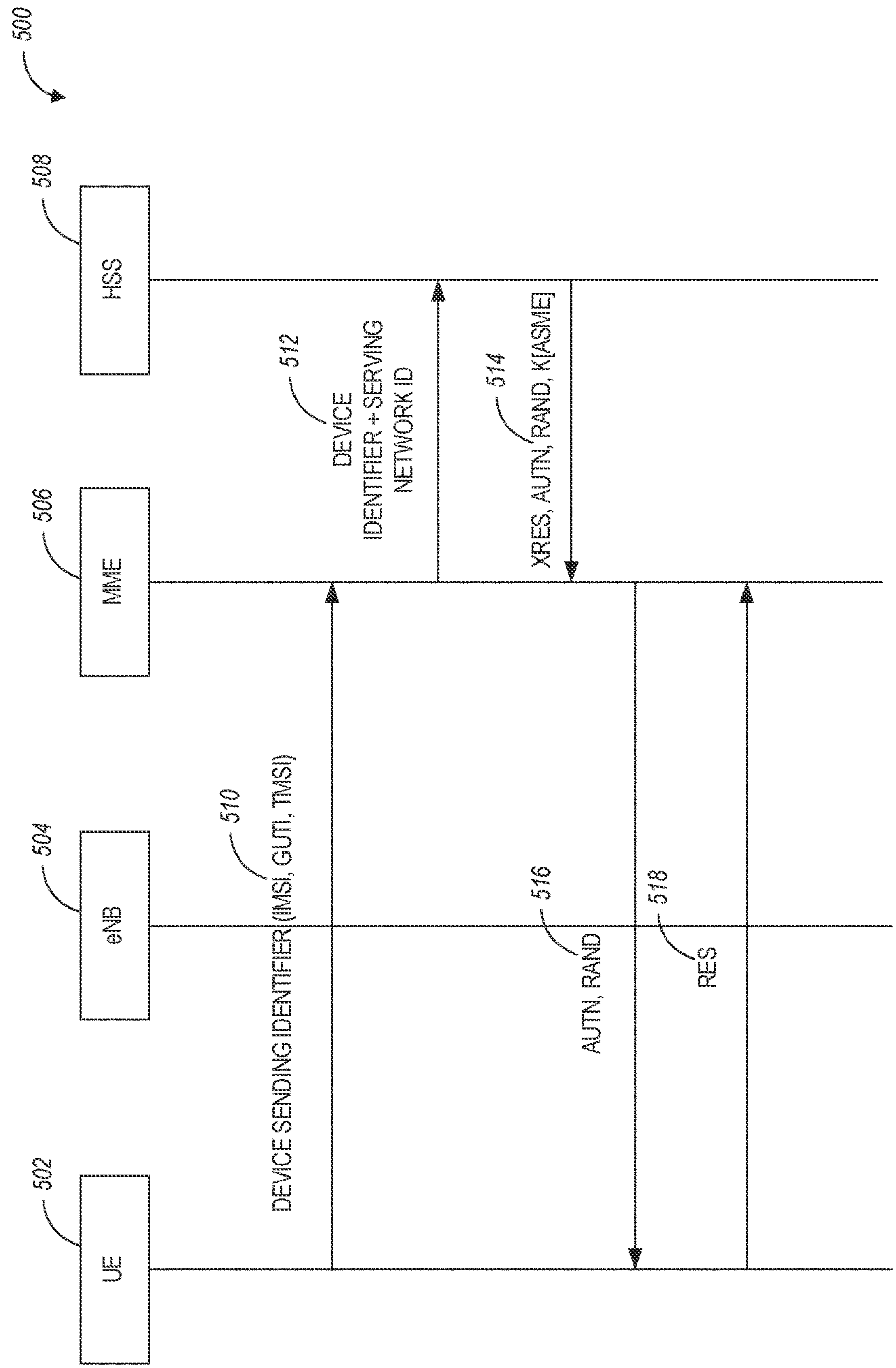

500
502
UE
504
eNB
506
MME
508
HSS
510
DEVICE SENDING IDENTIFIER (IMSI, GUTI, TMSI)
512
DEVICE IDENTIFIER + SERVING NETWORK ID
514
XRES, AUTN, RAND, K[ASME]
516
AUTN, RAND
518
RES

FIG. 5

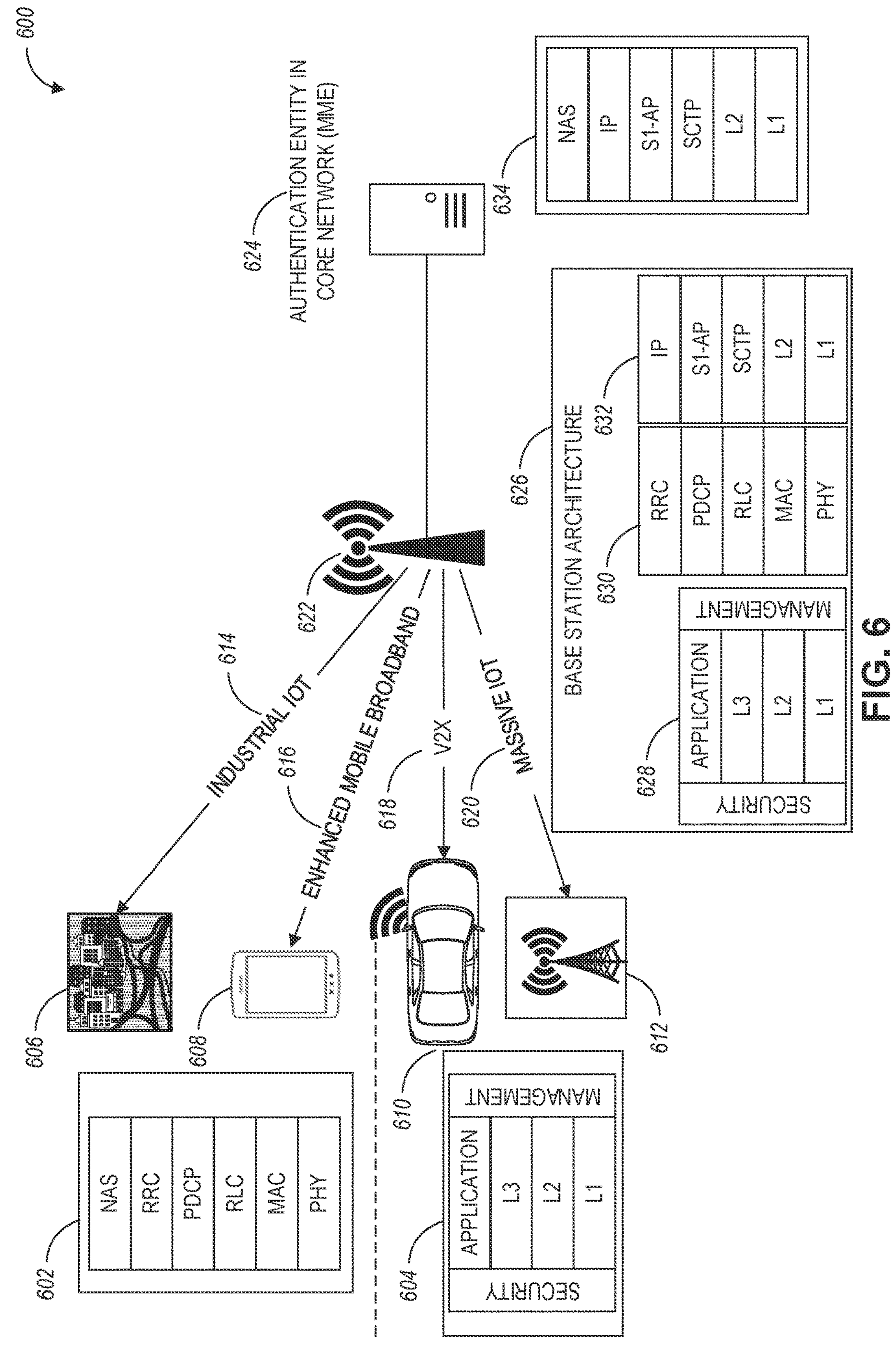

600
624
AUTHENTICATION ENTITY IN CORE NETWORK (MME)
634
NAS
IP
S1-AP
SCTP
L2
L1
622
614
INDUSTRIAL IOT
616
ENHANCED MOBILE BROADBAND
618
V2X
620
MASSIVE IOT
626
BASE STATION ARCHITECTURE
628
630
632
APPLICATION
L3
L2
L1
SECURITY
MANAGEMENT
RRC
PDCP
RLC
MAC
PHY
IP
S1-AP
SCTP
L2
L1
606
608
610
612
602
NAS
RRC
PDCP
RLC
MAC
PHY
604
APPLICATION
L3
L2
L1
SECURITY
MANAGEMENT

FIG. 6

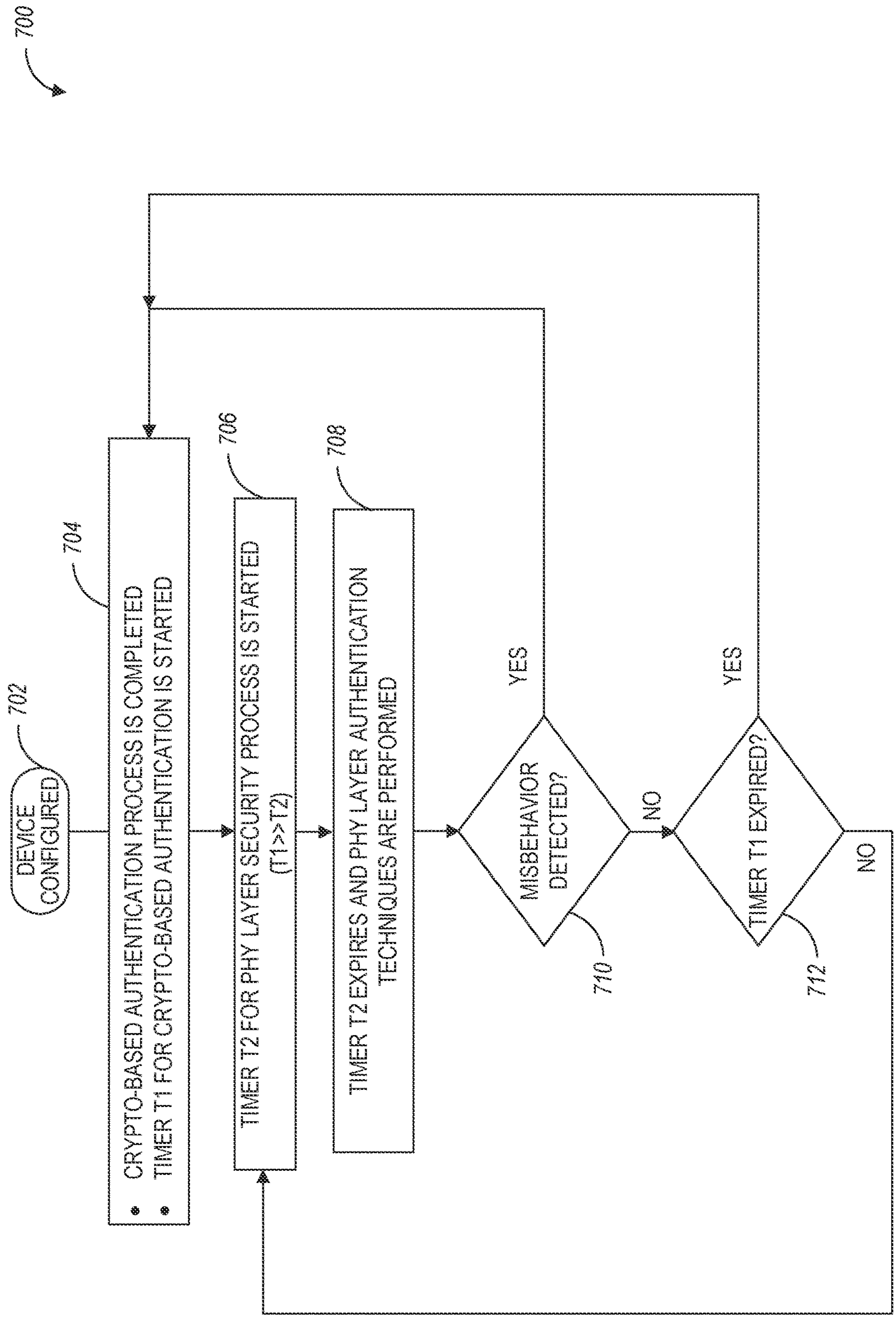

700
702
DEVICE CONFIGURED
704
• CRYPTO-BASED AUTHENTICATION PROCESS IS COMPLETED
• TIMER T1 FOR CRYPTO-BASED AUTHENTICATION IS STARTED
706
TIMER T2 FOR PHY LAYER SECURITY PROCESS IS STARTED (T1>>T2)
708
TIMER T2 EXPIRES AND PHY LAYER AUTHENTICATION TECHNIQUES ARE PERFORMED
710
MISBEHAVIOR DETECTED?
YES
NO
712
TIMER T1 EXPIRED?
YES
NO

FIG. 7

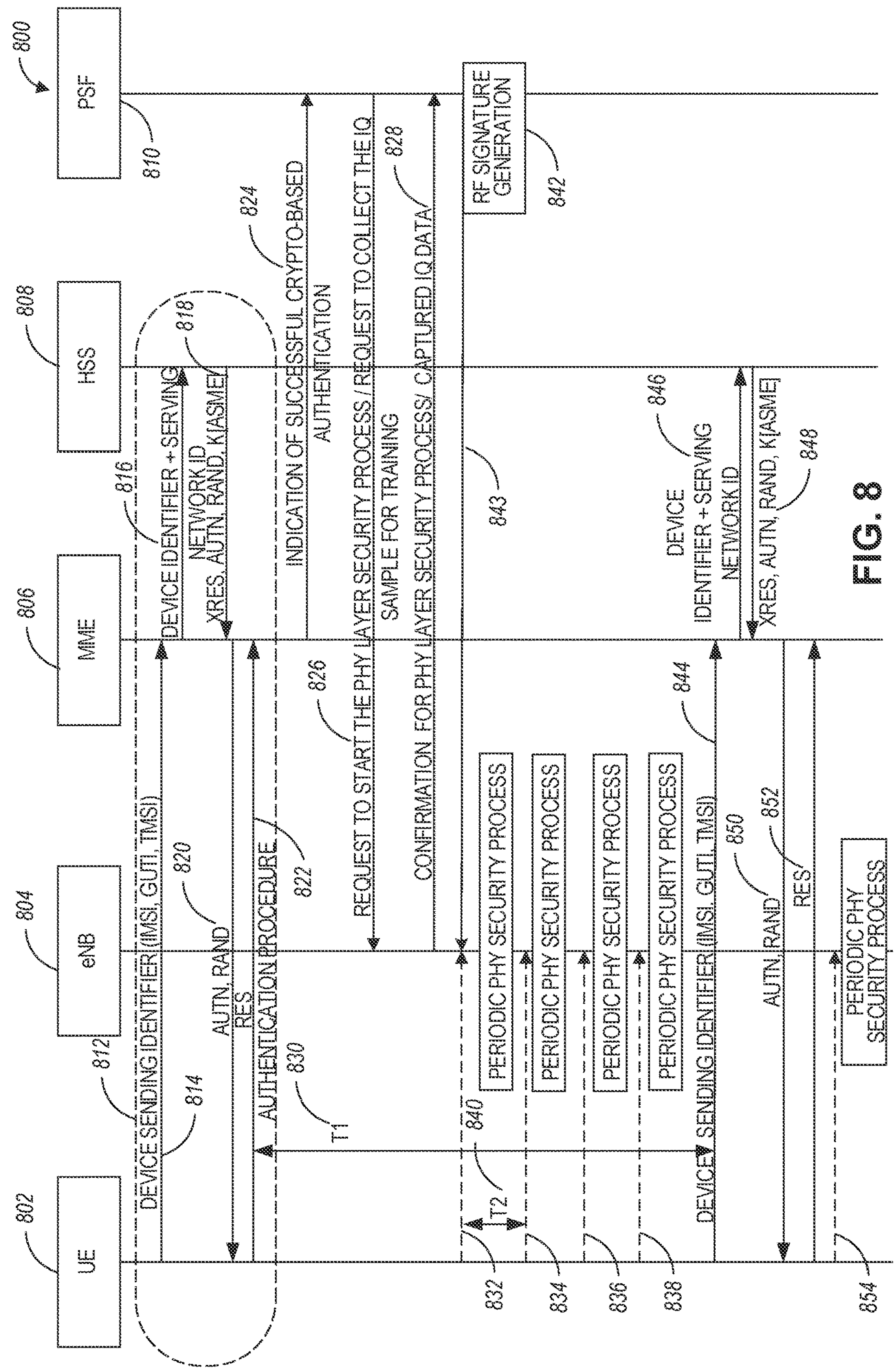

800
UE
802
eNB
804
MME
806
HSS
808
PSF
810
DEVICE SENDING IDENTIFIER (IMSI, GUTI, TMSI)
812
814
DEVICE IDENTIFIER + SERVING NETWORK ID
816
XRES, AUTN, RAND, K[ASME]
818
AUTN, RAND
820
RES
AUTHENTICATION PROCEDURE
822
INDICATION OF SUCCESSFUL CRYPTO-BASED AUTHENTICATION
824
REQUEST TO START THE PHY LAYER SECURITY PROCESS / REQUEST TO COLLECT THE IQ SAMPLE FOR TRAINING
826
CONFIRMATION FOR PHY LAYER SECURITY PROCESS/ CAPTURED IQ DATA
828
T1
830
RF SIGNATURE GENERATION
842
843
840
T2
PERIODIC PHY SECURITY PROCESS
PERIODIC PHY SECURITY PROCESS
PERIODIC PHY SECURITY PROCESS
PERIODIC PHY SECURITY PROCESS
832
834
836
838
DEVICE SENDING IDENTIFIER (IMSI, GUTI, TMSI)
844
DEVICE IDENTIFIER + SERVING NETWORK ID
846
XRES, AUTN, RAND, K[ASME]
848
AUTN, RAND
850
RES
852
PERIODIC PHY SECURITY PROCESS
854

FIG. 8

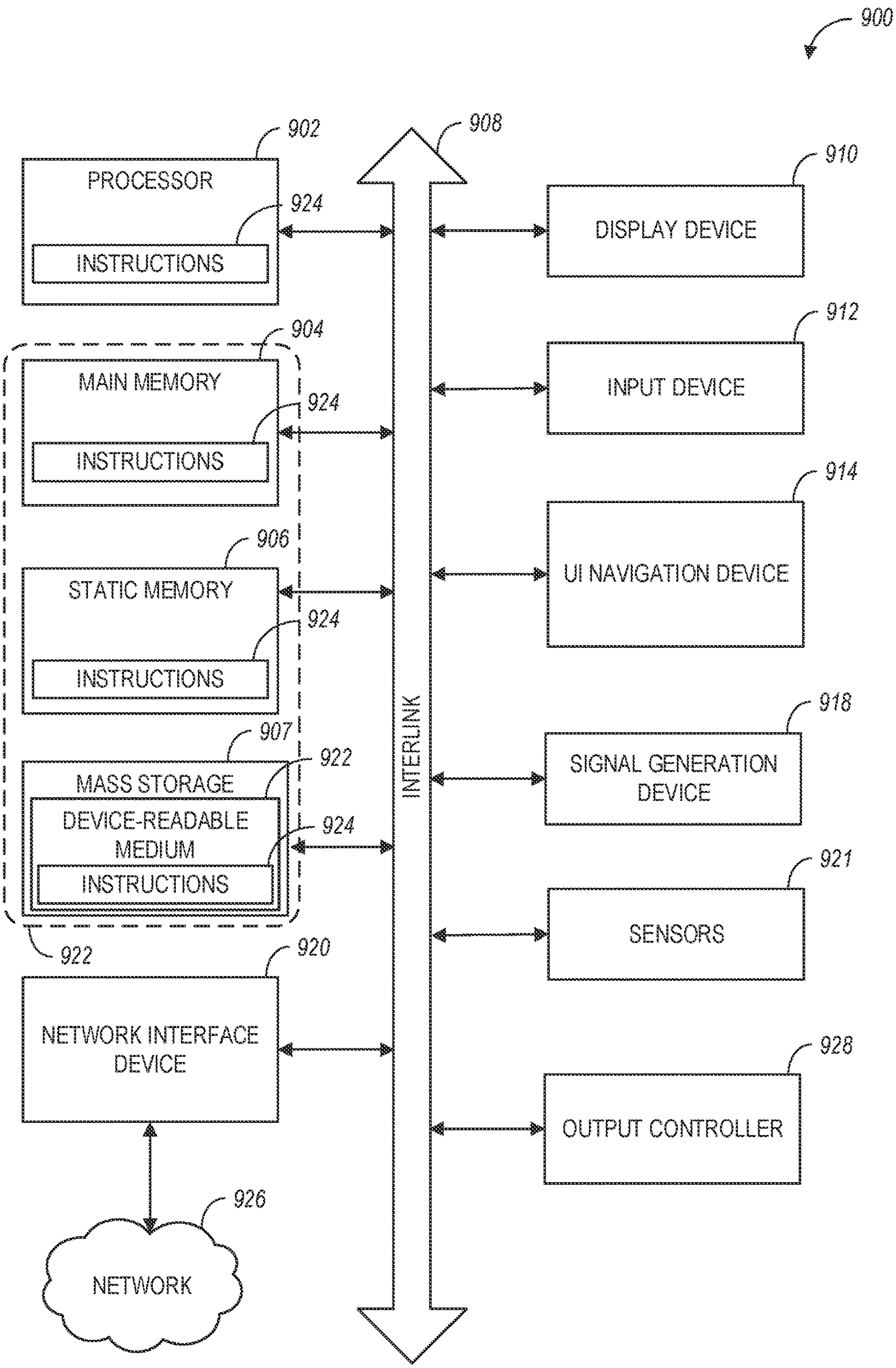

900
902
PROCESSOR
924
INSTRUCTIONS
904
MAIN MEMORY
924
INSTRUCTIONS
906
STATIC MEMORY
924
INSTRUCTIONS
907
MASS STORAGE
922
DEVICE-READABLE MEDIUM
924
INSTRUCTIONS
922
920
NETWORK INTERFACE DEVICE
926
NETWORK
908
INTERLINK
910
DISPLAY DEVICE
912
INPUT DEVICE
914
UI NAVIGATION DEVICE
918
SIGNAL GENERATION DEVICE
921
SENSORS
928
OUTPUT CONTROLLER

FIG. 9

ADAPTIVE AND HIERARCHICAL NETWORK AUTHENTICATION FRAMEWORK

TECHNICAL FIELD

Aspects pertain to wireless communications. Some aspects relate to wireless networks including 3GPP (Third Generation Partnership Project) networks. 3GPP LTE (Long Term Evolution) networks, 3GPP LTE-A (LTE Advanced) networks, (MulteFire, LTE-U), and fifth-generation (5G) networks including 5G new radio (NR) (or 5G-NR) networks, 5G-LTE networks such as 5G NR unlicensed spectrum (NR-U) networks and other unlicensed networks including Wi-Fi, CBRS (OnGo), etc. Other aspects are directed to adaptive and hierarchical network authentication framework for 5G and beyond networks.

BACKGROUND

Mobile communications have evolved significantly from early voice systems to today's highly sophisticated integrated communication platform. With the increase in different types of devices communicating with various network devices, usage of 3GPP systems has increased. The penetration of mobile devices (user equipment or UEs) in modern society has continued to drive demand for a wide variety of networked devices in many disparate environments. Fifth-generation (5G) and subsequent generations of wireless systems are forthcoming and are expected to enable even greater speed, connectivity, and usability. Next generation 5G networks (or NR networks) are expected to increase throughput, coverage, and robustness and reduce latency and operational and capital expenditures. 5G-NR networks will continue to evolve based on 3GPP LTE-Advanced with additional potential new radio access technologies (RATS) to enrich people's lives with seamless wireless connectivity solutions delivering fast, rich content and services. As current cellular network frequency is saturated, higher frequencies, such as millimeter wave (mmWave) frequency, can be beneficial due to their high bandwidth.

Potential LTE operation in the unlicensed spectrum includes (and is not limited to) the LTE operation in the unlicensed spectrum via dual connectivity (DC), or DC-based LAA, and the standalone LTE system in the unlicensed spectrum, according to which LTE-based technology solely operates in the unlicensed spectrum without requiting an "anchor" in the licensed spectrum, called MulteFire. MulteFire combines the performance benefits of LTE technology with the simplicity of Wi-Fi-like deployments.

Further enhanced operation of LTE and NR systems in the licensed, as well as unlicensed spectrum, is expected in future releases and 5G (and beyond) communication systems. Such enhanced operations can include techniques for adaptive and hierarchical network authentication framework for 5G and beyond networks.

BRIEF DESCRIPTION OF THE FIGURES

In the figures, which are not necessarily drawn to scale, like numerals may describe similar components in different views. Like numerals having different letter suffixes may represent different instances of similar components. The figures illustrate generally, by way of example, but not by way of limitation, various aspects discussed in the present document.

FIG. **1**A illustrates an exemplary architecture of a network, in accordance with some aspects.

FIG. **1**B and FIG. **1**C illustrate a non-roaming 5G system architecture in accordance with some aspects.

FIG. **2**, FIG. **3**, and FIG. **4** illustrate various systems, devices, and components that may implement aspects of disclosed embodiments.

FIG. **5** illustrates a cryptography-based network authentication framework, according to some example embodiments.

FIG. **6** illustrates an example architecture of a 5G and beyond network, according to some example embodiments.

FIG. **7** illustrates a flow chart of functionalities performed in a hierarchical authentication framework, according to some example embodiments.

FIG. **8** illustrates a swimlane diagram of example communications flow in a hierarchical authentication framework, according to some example embodiments.

FIG. **9** illustrates a block diagram of a communication device such as an evolved Node-B (eNB), a new generation Node-B (gNB) (or another RAN node), an access point (AP), a wireless station (STA), a mobile station (MS), or a user equipment (UE), in accordance with some aspects.

DETAILED DESCRIPTION

The following description and the drawings sufficiently illustrate aspects to enable those skilled in the art to practice them. Other aspects may incorporate structural, logical, electrical, process, and other changes. Portions and features of some aspects may be included in or substituted for, those of other aspects. Aspects outlined in the claims encompass all available equivalents of those claims.

FIG. **1**A illustrates an architecture of a network in accordance with some aspects. The network **140**A is shown to include user equipment (UE) **101** and UE **102**. The UEs **101** and **102** are illustrated as smartphones (e.g., handheld touchscreen mobile computing devices connectable to one or more cellular networks) but may also include any mobile or non-mobile computing device, such as Personal Data Assistants (PDAs), pagers, laptop computers, desktop computers, wireless handsets, drones, or any other computing device including a wired and/or wireless communications interface. The UEs **101** and **102** can be collectively referred to herein as UE **101**, and UE **101** can be used to perform one or more of the techniques disclosed herein.

Any of the radio links described herein (e.g., as used in the network **140**A or any other illustrated network) may operate according to any exemplary radio communication technology and/or standard.

LTE and LTE-Advanced are standards for wireless communications of high-speed data for UE such as mobile telephones. In LTE-Advanced and various wireless systems, carrier aggregation is a technology according to which multiple carrier signals operating on different frequencies may be used to carry communications for a single UE, thus increasing the bandwidth available to a single device. In some aspects, carrier aggregation may be used where one or more component carriers operate on unlicensed frequencies.

Aspects described herein can be used in the context of any spectrum management scheme including, for example, dedicated licensed spectrum, unlicensed spectrum. (licensed) shared spectrum (such as Licensed Shared Access (LSA) in 2.3-2.4 GHz, 3.4-3.6 GHz, 3.6-3.8 GHz, and further frequencies and Spectrum Access System (SAS) in 3.55-3.7 GHz and further frequencies).

Aspects described herein can also be applied to different Single Carrier or OFDM flavors (CP-OFDM, SC-FDMA, SC-OFDM, filter bank-based multicarrier (FBMC), OFDMA, etc.) and in particular 3GPP NR (New Radio) by allocating the OFDM carrier data bit vectors to the corresponding symbol resources.

In some aspects, any of the UEs **101** and **102** can comprise an Internet-of-Things (IoT) UE or a Cellular IoT (CIoT) UE, which can comprise a network access layer designed for low-power IoT applications utilizing short-lived UE connections. In some aspects, any of the UEs **101** and **102** can include a narrowband (NB) IoT UE (e.g., such as an enhanced NB-IoT (eNB-IoT) UE and Further Enhanced (FeNB-IoT) UE). An IoT UE can utilize technologies such as machine-to-machine (M2M) or machine-type communications (MTC) for exchanging data with an MTC server or device via a public land mobile network (PLMN), Proximity-Based Service (ProSe), or device-to-device (D2D) communication, sensor networks, or networks. The M2M or MTC exchange of data may be a machine-initiated exchange of data. An IoT network includes interconnecting IoT UEs, which may include uniquely identifiable embedded computing devices (within the Internet infrastructure), with short-lived connections. The IoT UEs may execute background applications (e.g., keep-alive messages, status updates, etc.) to facilitate the connections of the IoT network.

In some aspects, any of the UEs **101** and **102** can include enhanced MTC (eMTC) UEs or further enhanced MTC (FeMTC) UEs.

The UEs **101** and **102** may be configured to connect, e.g., communicatively couple, with a radio access network (RAN) **110**. The RAN **110** may be, for example, a Universal Mobile Telecommunications System (UMTS), an Evolved Universal Terrestrial Radio Access Network (E-UTRAN), a NextGen RAN (NG RAN), or some other type of RAN. The UEs **101** and **102** utilize connections **103** and **104**, respectively, each of which comprises a physical communications interface or layer (discussed in further detail below); in this example, the connections **103** and **104** are illustrated as an air interface to enable communicative coupling and can be consistent with cellular communications protocols, such as a Global System for Mobile Communications (GSM) protocol, a code-division multiple access (CDMA) network protocol, a Push-to-Talk (PTT) protocol, a PTT over Cellular (POC) protocol, a Universal Mobile Telecommunications System (UMTS) protocol, a 3GPP Long Term Evolution (LTE) protocol, a filth-generation (5G) protocol, a New Radio (NR) protocol, and the like.

In an aspect, the UEs **101** and **102** may further directly exchange communication data via a ProSe interface **105**. The ProSe interface **105** may alternatively be referred to as a sidelink interface comprising one or more logical channels, including but not limited to a Physical Sidelink Control Channel (PSCCH), a Physical Sidelink Shared Channel (PSSCH), a Physical Sidetink Discovery Channel (PSDCH), and a Physical Sidelink Broadcast Channel (PSBCH).

The UE **102** is shown to be configured to access an access point (AP) **106** via connection **107**. The connection **107** can comprise a local wireless connection, such as, for example, a connection consistent with any IEEE 802.11 protocol, according to which the AP **106** can comprise a wireless fidelity (WiFi®) router. In this example, the AP **106** is shown to be connected to the Internet without connecting to the core network of the wireless system (described in further detail below).

The RAN **110** can include one or more access nodes that enable connections **103** and **104**. These access nodes (ANs) can be referred to as base stations (BSs), NodeBs, evolved NodeBs (eNBs), Next Generation NodeBs (gNBs), RAN network nodes, and the like, and can comprise ground stations (e.g., terrestrial access points) or satellite stations providing coverage within a geographic area (e.g., a cell). In some aspects, the communication nodes **111** and **112** can be transmission/reception points (TRPs). In instances when the communication nodes **111** and **112** are NodeBs (e.g., eNBs or gNBs), one or more TRPs can function within the communication cell of the NodeBs. The RAN **110** may include one or more RAN nodes for providing macrocells, e.g., macro RAN node **111**, and one or more RAN nodes for providing femtocells or picocells (e.g., cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells), e.g., low power (LP) RAN node **112** or an unlicensed spectrum based secondary RAN node **112**.

Any of the RAN nodes **111** and **112** can terminate the air interface protocol and can be the first point of contact for the UEs **101** and **102**. In some aspects, any of the RAN nodes **111** and **112** can fulfill various logical functions for the RAN **110** including, but not limited to, radio network controller (RNC) functions such as radio bearer management, uplink and downlink dynamic radio resource management, and data packet scheduling, and mobility management. In an example, any of the nodes **111** and/or **112** can be a new generation Node-B (gNB), an evolved node-B (eNB), or another type of RAN node.

The RAN **110** is shown to be communicatively coupled to a core network (CN) **120** via an S1 interface **113**. In aspects, the CN **120** may be an evolved packet core (EPC) network, a NextGen Packet Core (NPC) network, or some other type of CN (e.g., as illustrated in reference to FIGS. **1B-1C**). In this aspect, the S1 interface **113** is split into two parts: the S1-U interface **114**, which carries user traffic data between the RAN nodes **111** and **112** and the serving gateway (S-GW) **122**, and the S1-mobility management entity (MME) interface **115**, which is a signaling interface between the RAN nodes **111** and **112** and MMEs **121**.

In this aspect, the CN **120** comprises the MMEs **121**, the S-GW **122**, the Packet Data Network (PDN) Gateway (P-GW) **123**, and a home subscriber server (HSS) **124**. The MMEs **121** may be similar in function to the control plane of legacy Serving General Packet Radio Service (GPRS) Support Nodes (SGSN). The MMEs **121** may manage mobility aspects in access such as gateway selection and tracking area list management. The HSS **124** may comprise a database for network users, including subscription-related information to support the network entities' handling of communication sessions. The CN **120** may comprise one or several HSSs **124**, depending on the number of mobile subscribers, on the capacity of the equipment, on the organization of the network, etc. For example, the HSS **124** can provide support for routing/roaming, authentication, authorization, naming/addressing resolution, location dependencies, etc.

The S-GW **122** may terminate the S1 interface **113** towards the RAN **110**, and route data packets between the RAN **110** and the CN **120**. In addition, the S-GW **122** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities of the S-GW **122** may include lawful intercept, charging, and some policy enforcement.

The P-GW **123** may terminate an SGi interface toward a PDN. The P-GW **123** may route data packets between the EPC network **120** and external networks such as a network including the application server **184** (alternatively referred to as application function (AF)) via an Internet Protocol (IP) interface **125**. The P-GW **123** can also communicate data to other external networks **131**A, which can include the Internet, IP multimedia subsystem (IPS) network, and other networks. Generally, the application server **184** may be an element offering applications that use IP bearer resources with the core network (e.g., UMTS Packet Services (PS) domain, LTE PS data services, etc.). In this aspect, the P-GW **123** is shown to be communicatively coupled to an application server **184** via an IP interface **125**. The application server **184** can also be configured to support one or more communication services (e.g., Voice-over-Internet Protocol (VoIP) sessions, PTT sessions, group communication sessions, social networking services, etc.) for the UEs **101** and **102** via the CN **120**.

The P-GW **123** may further be a node for policy enforcement and charging data collection. Policy and Charging Rules Function (PCRF) **126** is the policy and charging control element of the CN **120**. In a non-roaming scenario, in some aspects, there may be a single PCRF in the Home Public Land Mobile Network (HPLMN) associated with a UE's Internet Protocol Connectivity Access Network (IP-CAN) session. In a roaming scenario with a local breakout of traffic, there may be two PCRFs associated with a UE's IP-CAN session: a Home PCRF (H-PCRF) within an HPLMN and a Visited PCRF (V-PCRF) within a Visited Public Land Mobile Network (VPLMN). The PCRF **126** may be communicatively coupled to the application server **184** via the P-GW **123**.

In some aspects, the communication network **140**A can be an IoT network or a 5G network, including a 5G new radio network using communications in the licensed (5G NR) and the unlicensed (5G NR-U) spectrum. One of the current enablers of IoT is the narrowband-IoT (NB-IoT).

An NG system architecture can include the RAN **110** and a 5G network core (5GC) **120**. The NG-RAN **110** can include a plurality of nodes, such as gNBs and NG-eNBs. The core network **120** (e.g., a 5G core network or 5GC) can include an access and mobility function (AMF) and/or a user plane function (UPF). The AMF and the UPF can be communicatively coupled to the gNBs and the NG-eNBs via NG interfaces. More specifically, in some aspects, the gNBs and the NG-eNBs can be connected to the AMF by NG-C interfaces, and to the UPF by NG-U interfaces. The gNBs and the NG-eNBs can be coupled to each other via Xn interfaces.

In some aspects, the NG system architecture can use reference points between various nodes as provided by 3GPP Technical Specification (TS) 23.501 (e.g., V15.4.0, 2018-12). In some aspects, each of the gNBs and the NG-eNBs can be implemented as a base station, a mobile edge server, a small cell, a home eNB, a RAN network node, and so forth. In some aspects, a gNB can be a master node (MN) and NG-eNB can be a secondary node (SN) in a 5G architecture. In some aspects, the master/primary node may operate in a licensed band and the secondary node may operate in an unlicensed band.

FIG. **1**B illustrates a non-roaming 5G system architecture in accordance with some aspects. Referring to FIG. **1**B, there is illustrated a 5G system architecture **140**B in a reference point representation. More specifically, UE **102** can be in communication with RAN **110** as well as one or more other 5G core (5GC) network entities. The 5G system architecture **140**B includes a plurality of network functions (INFs), such as access and mobility management function (AMF) **132**, session management function (SMF) **136**, policy control function (PCF) **148**, application function (AF) **150**, user plane function (UPF) **134**, network slice selection function (NSSF) **142**, authentication server function (AUSF) **144**, and unified data management (UDM)/home subscriber server (HSS) **146**. The UPF **134** can provide a connection to a data network (DN) **152**, which can include, for example, operator services, Internet access, or third-party services. The AMF **132** can be used to manage access control and mobility and can also include network slice selection functionality. The SMF **136** can be configured to set up and manage various sessions according to network policy. The UPF **134** can be deployed in one or more configurations according to the desired service type. The PCF **148** can be configured to provide a policy framework using network slicing, mobility management, and roaming (similar to in a 4G communication system). The UDM can be configured to store subscriber profiles and data (similar to an HSS in a 4G communication system).

In some aspects, the 5G system architecture **140**B includes an IP multimedia subsystem (IMS) **168**B as well as a plurality of IP multimedia core network subsystem entities, such as call session control functions (CSCFs). More specifically, the IMS **168**B includes a CSCF, which can act as a proxy CSCF (P-CSCF) **162**BE, a serving CSCF (S-CSCF) **164**B, an emergency CSCF (E-CSCF) (not illustrated in FIG. **1**B), or interrogating; CSCF (I-CSCF) **166**B. The P-CSCF **162**B can be configured to be the first contact point for the UE **102** within the IM subsystem (IMS) **168**B. The S-CSCF **164**B can be configured to handle the session states in the network, and the E-CSCF can be configured to handle certain aspects of emergency sessions such as routing an emergency request to the correct emergency center or PSAP. The I-CSCF **166**B can be configured to function as the contact point within an operator's network for all IMS connections destined to a subscriber of that network operator, or a roaming subscriber currently located within that network operator's service area. In some aspects, the I-CSCF **166**B can be connected to another IP multimedia network **170**E, e.g. an IMS operated by a different network operator.

In some aspects, the UDM/HSS **146** can be coupled to an application server **160**E, which can include a telephony application server (TAS) or another application server (AS). The AS **160**B can be coupled to the BIS **168**B via the S-CSCF **164**B or the I-CSCF **166**B.

A reference point representation shows that interaction can exist between corresponding NF services. For example, FIG. **1**B illustrates the following reference points: N1 (between the UE **102** and the AMF **132**), N2 (between the RAN **110** and the AMF **132**), N3 (between the RAN **110** and the UPF **134**), N4 (between the SMF **136** and the UPF **134**), N5 (between the PCF **148** and the AF **150**, not shown), N6 (between the UPF **134** and the DN **152**), N7 (between the SMF **136** and the PCF **148**, not shown), N8 (between the UDM **146** and the AMF **132**, not shown), N9 (between two UPFs **134**, not shown), N**10** (between the UDM **146** and the SMF **136**, not shown), N**11** (between the AMF **132** and the SMF **136**, not shown), N**12** (between the AUSF **144** and the AMF **132**, not shown), N**13** (between the AUSF **144** and the UDM **146**, not shown), N**14** (between two AM's **132**, not shown), N**15** (between the PCF **148** and the AMF **132** in case of a non-roaming scenario, or between the PCF **148** and a visited network and AMF **132** in case of a roaming scenario, not shown), N**16** (between two SMFs, not shown), and N22 (between AMF **132** and NSSF **142**, not shown). Other reference point representations not shown in FIG. **1**B can also be used.

FIG. **1**C illustrates a 5G system architecture **140**C and a service-based representation. In addition to the network entities illustrated in FIG. **1**B, system architecture **140**C can also include a network exposure function (NEF) **154** and a network repository function (NRF) **156**. In some aspects, 5G system architectures can be service-based and interaction between network functions can be represented by corresponding point-to-point reference points Ni or as service-based interfaces.

In some aspects, as illustrated in FIG. **1**C, service-based representations can be used to represent network functions within the control plane that enable other authorized network functions to access their services. In this regard, 5G system architecture **140**C can include the following service-based interfaces: Namf **158**H (a service-based interface exhibited by the AMF **132**), Nsmf **158**I (a service-based interface exhibited by the SMF **136**), Nnef **158**B (a service-based interface exhibited by the NEF **154**), Npcf **158**D (a service-based interface exhibited by the PCF **148**), a Nudm **158**E (a service-based interface exhibited by the UDM **146**), Naf **1558**F (a service-based interface exhibited by the AF **150**), Nnrf **158**C (a service-based interface exhibited by the NRF **156**), Nnssf **158**A (a service-based interface exhibited by the NSSF **142**), Nausf **158**G (a service-based interface exhibited by the AUSF **144**). Other service-based interfaces (e.g., Nudr, N5g-eir, and Nudsf) not shown in FIG. **1**C can also be used.

FIG. **2**, FIG. **3**, and FIG. **4** illustrate various systems, devices, and components that may implement aspects of disclosed embodiments. More specifically, UEs and/or base stations (such as gNBs) discussed in connection with FIGS. **1**A-**4** can be configured to perform the disclosed techniques.

FIG. **2** illustrates a network **200** in accordance with various embodiments. The network **200** may operate in a manner consistent with 3GPP technical specifications for LTE or 5G/NR systems. However, the example embodiments are not limited in this regard and the described embodiments may apply to other networks that benefit from the principles described herein, such as future 3GPP systems, or the like.

The network **200** may include a UE **202**, which may include any mobile or non-mobile computing device designed to communicate with a RAN **204** via an over-the-air connection. The UE **202** may be, but is not limited to, a smartphone, tablet computer, wearable computing device, desktop computer, laptop computer, in-vehicle infotainment, in-car entertainment device, instrument cluster, head-up display device, onboard diagnostic device, dashtop mobile equipment, mobile data terminal, electronic engine management system, electronic/engine control unit, electronic/engine control module, embedded system, sensor, microcontroller, control module, engine management system, networked appliance, machine-type communication device, M2M or D2D device, IoT device, etc.

In some embodiments, the network **200** may include a plurality of UEs coupled directly with one another via a sidelink interface. The UEs may be M2M/D2M) devices that communicate using physical sidelink channels such as but not limited to, PSBCH, PSDCH, PSSCH, PSCCH, PSFCH, etc.

In some embodiments, the UE **202** may additionally communicate with an AP **206** via an over-the-air connection. The AP **206** may manage a WLAN connection, which may serve to offload some/all network traffic from the RAN **204**. The connection between the UE **202** and the AP **206** may be consistent with any IEEE 802.11 protocol, wherein the AP **206** could be a wireless fidelity (Wi-Fi) router. In some embodiments, the UE **202**, RAN **204**, and AP **206** may utilize cellular-WLAN aggregation (for example, LWA/LWIP), Cellular-WLAN aggregation may involve the UE **202** being configured by the RAN **204** to utilize both cellular radio resources and WLAN resources.

The RAN **204** may include one or more access nodes, for example, access node (AN) **208**. AN **208** may terminate air-interface protocols for the UE **202** by providing access stratum protocols including RRC, Packet Data Convergence Protocol (PDCP), Radio Link Control (RLC), MAC, and L1 protocols. In this manner, the AN **208** may enable data/voice connectivity between the core network (CN) **220** and the UE **202**. In some embodiments, the AN **208** may be implemented in a discrete device or as one or more software entities running on server computers as part of, for example, a virtual network, which may be referred to as a CRAN or virtual baseband unit pool. The AN **208** be referred to as a BS, gNB, RAN node, eNB, ng-eNB, NodeB, RSU, TRxP, TRP, etc. The AN **208** may be a macrocell base station or a low-power base station for providing femtocells, picocells, or other like cells having smaller coverage areas, smaller user capacity, or higher bandwidth compared to macrocells.

In embodiments in which the RAN **204** includes a plurality of ANs, they may be coupled with one another via an X2 interface (if the RAN **204** is an LTE RAN) or an Xn interface (if the RAN **204** is a 5G RAN). The X2/Xn interfaces, which may be separated into control/user plane interfaces in some embodiments, may allow the ANs to communicate information related to handovers, data/context transfers, mobility, load management, interference coordination, etc.

The ANs of the RAN **204** may each manage one or more cells, cell groups, component carriers, etc. to provide the UE **202** with an air interface for network access. The UE **202** may be simultaneously connected with a plurality of cells provided by the same or different ANs of the RAN **204**. For example, the UE **202** and RAN **204** may use carrier aggregation to allow the UE **202** to connect with a plurality of component carriers, each corresponding to a Pcell or Scell. In dual connectivity scenarios, a first AN may be a master node that provides an MCG and a second AN may be a secondary node that provides an SCG. The first/second ANs may be any combination of eNB, gNB, ng-eNB, etc.

The RAN **204** may provide the air interface over a licensed spectrum or an unlicensed spectrum. To operate in the unlicensed spectrum, the nodes may use LAA, eLAA, and/or feLAA mechanisms based on CA technology with PCells/Scells. Before accessing the unlicensed spectrum, the nodes may perform medium/carrier-sensing operations based on, for example, a listen-before-talk (LBT) protocol.

In V2X scenarios, the UE **202** or AN **208** may be or act as a roadside unit (RSU), which may refer to any transportation infrastructure entity used for V2X communications. An RSU may be implemented in or by a suitable AN or a stationary (or relatively stationary) UE. An RSU implemented in or by: a UE may be referred to as a "UE-type RSU"; an eNB may be referred to as an "eNB-type RSU"; a gNB may be referred to as a "gNB-type RSU"; and the like. In one example, an RSU is a computing device coupled with radio frequency circuitry located on a roadside that provides connectivity support to passing vehicle UEs. The RSU may also include internal data storage circuitry to store intersection map geometry, traffic statistics, media, as well as applications/software to sense and control ongoing vehicular and pedestrian traffic. The RSU may provide very low latency communications required for high-speed events, such as crash avoidance, traffic warnings, and the like.

Additionally, or alternatively, the RSU may provide other cellular/WLAN communications services. The components of the RSU may be packaged in a weatherproof enclosure suitable for outdoor installation and may include a network interface controller to provide a wired connection (e.g., Ethernet) to a traffic signal controller or a backhaul network.

In some embodiments, the RAN **204** may be an LTE RAN **210** with eNBs, for example, eNB **212**. The LTE RAN **210** may provide an LTE air interface with the following characteristics: sub-carrier spacing (SCS) of 15 kHz; CP-OFDM waveform for downlink (DL) and SC-FDMA waveform for uplink (UL); turbo codes for data and TBCC for control; etc. The LTE air interface may rely on CSI-RS for CSI acquisition and beam management; PDSCH/PDCCH DMARS for PDSCH/PDCCH demodulation; and CRS for cell search and initial acquisition, channel quality measurements, and channel estimation for coherent demodulation/detection at the UE. The LTE air interface may operate on sub-6 GHz bands.

In some embodiments, the RAN **204** may be an NG-RAN **214** with gNBs, for example, gNB **216**, or ng-eNBs, for example, ng-eNB **218**. The gNB **216** may connect with 5G-enabled UEs using a 5G NR interface. The gNB **216** may connect with a 5G core through an NG interface, which may include an N2 interface or an N3 interface. The ng-eNB **218** may also connect with the 5G core through an NG interface but may connect with a UE via an LTE air interface. The gNB **216** and the ng-eNB **218** may connect over an Xn interface.

In some embodiments, the NG interface may be split into two parts, an NG user plane (NG-U) interface, which carries traffic data between the nodes of the NG-RAN **214** and a UPF **248** (e.g., N3 interface), and an NG control plane (NG-C) interface, which is a signaling interface between the nodes of the NG-RAN **214** and an AMF **244** (e.g., N2 interface).

The NG-RAN **214** may provide a 5G-NR air interface with the following characteristics: variable SCS; CP-OFDM for DL, CP-OFDM and DFT-s-OFDM for UL; polar, repetition, simplex, and Reed-Muller codes for control and LDPC for data. The 5G-NR air interface may rely on CSI-RS, PDSCH/PDCCH DMRS similar to the LTE air interface. The 5G-NR air interface may not use a CRS but may use PBCH DMRS for PBCH demodulation; PTRS for phase tracking for PDSCH and tracking reference signal for time tracking. The 5G-NR air interface may operate on FR1 bands that include sub-6 GHz bands or FR2 bands that include bands from 24.25 GHz to 52.6 GHz. The 5G-NR air interface may include a synchronization signal and physical broadcast channel (SS/PBCH) block (SSB) that is an area of a downlink resource grid that includes PSS/SSS/PBCH.

In some embodiments, the 5G-NR air interface may utilize BWPs (bandwidth parts) for various purposes. For example, BWP can be used for dynamic adaptation of the SCS. For example, the UE **202** can be configured with multiple BWPs where each BWP configuration has a different SCS. When a BWP change is indicated to the UE **202**, the SCS of the transmission is changed as well. Another use case example of BWP is related to power saving. In particular, multiple BWPs can be configured for the UE **202** with different amounts of frequency resources (for example, PRBs) to support data transmission under different traffic loading scenarios. A BWP containing a smaller number of PRBs can be used for data transmission with a small traffic load while allowing power saving at the UE **202** and in some cases at the gNB **216**. A BWP containing a larger number of PRBs can be used for scenarios with higher traffic loads.

The RAN **204** is communicatively coupled to CN **220** that includes network elements to provide various functions to support data and telecommunications services to customers/subscribers (for example, users of UE **202**). The components of the CN **220** may be implemented in one physical node or separate physical nodes. In some embodiments, NFV may be utilized to virtualize any or all of the functions provided by the network elements of the CN **220** onto physical compute/storage resources in servers, switches, etc. A logical instantiation of the CN **220** may be referred to as a network slice, and a logical instantiation of a portion of the CN **220** may be referred to as a network sub-slice.

In some embodiments, the CN **220** may be connected to the LTE radio network as part of the Enhanced Packet System (EPS) **222**, which may also be referred to as an EPC (or enhanced packet core). The EPC **222** may include MME **224**, SGW **226**, SGSN **228**, HSS **230**, PGW **232**, and PCRF **234** coupled with one another over interfaces (or "reference points") as shown. Functions of the elements of the EPC **222** may be briefly introduced as follows.

The MIME **224** may implement mobility management functions to track the current location of the UE **202** to facilitate paging, bearer activation/deactivation, handovers, gateway selection, authentication, etc.

The SGW **226** may terminate an S1 interface toward the RAN and route data packets between the RAN and the EPC **222**. The SGW **226** may be a local mobility anchor point for inter-RAN node handovers and also may provide an anchor for inter-3GPP mobility. Other responsibilities may include lawful intercept, charging, and some policy enforcement.

The SGSN **228** may track the location of the UE **202** and perform security functions and access control. In addition, the SGSN **228** may perform inter-EPC node signaling for mobility between different RAT networks; PDN and S-GW selection as specified by MME **224**; MME selection for handovers; etc. The S3 reference point between the MME **221** and the SGSN **228** may enable user and bearer information exchange for inter-3GPP access network mobility in idle/active states.

The HSS **230** may include a database for network users, including subscription-related information to support the network entities' handling of communication sessions. The HSS **230** can provide support for routing/roaming, authentication, authorization, naming addressing resolution, location dependencies, etc. An S6a reference point between the HSS **230** and the MME **224** may enable the transfer of subscription and authentication data for authenticating/authorizing user access to the LTE CN **220**.

The PGW **232** may terminate an SGi interface toward a data network (DN) **236** that may include an application/content server **238**. The PGW **232** may route data packets between the LTE CN **220** and the data network **236**. The PGW **232** may be coupled with the SGW **226** by an S5 reference point to facilitate user plane tunneling and tunnel management. The **232** may further include a node for policy enforcement and charging data collection (for example, PCEF). Additionally, the SGi reference point between the PGW **232** and the data network **236** may be an operator external public, a private PDN, or an intra-operator packet data network, for example, for provision of IMS services. The PGW **232** may be coupled with a PCRF **234** via, a Gx reference point.

The PCRF **234** is the policy and charging control element of the CN **220**. The PCRF **234** may be communicatively coupled to the app/content server **238** to determine appropriate QoS and charging parameters for service flows. The

PCRF **234** may provision associated rules into a PCEF (via Gx reference point) with appropriate TFT and QCI.

In some embodiments, the CN **220** may be a 5GC **240**. The 5GC **240** may include an AUSF **242**, AMF **244**, SMF **246**, UPF **248**, NSSF **250**, NEF **252**, NRF **254**, PCF **256**, UDM **258**, and AF **260** coupled with one another over interfaces (or "reference points") as shown. Functions of the elements of the 5GC **240** may be briefly introduced as follows.

The AUSF **242** may store data for authentication of UE **202** and handle authentication-related functionality. The AUSF **242** may facilitate a common authentication framework for various access types. In addition to communicating with other elements of the 5GC **240** over reference points as shown, the AUSF **242** may exhibit a Nausf service-based interface.

The AMF **244** may allow other functions of the 5GC **240** to communicate with the UE **202** and the RAN **204** and to subscribe to notifications about mobility events with respect to the UE **202**. The AMF **244** may be responsible for registration management (for example, for registering UE **202**), connection management, reachability management, mobility management, lawful interception of AMF-related events, and access authentication and authorization. The AMF **244** may provide transport for SM messages between the UE **202** and the SWF **246**, and act as a transparent proxy for routing SM messages. AMF **244** may also provide transport for SMS messages between UE **202** and an SMSF. AMF **244** may interact with the AUSF **242** and the UE **202** to perform various security anchor and context management functions. Furthermore, AMF **244** may be a termination point of a RAN CP interface, which may include or be an N2 reference point between the RAN **204** and the AMF **244**; and the AMF **244** may be a termination point of NAS (N1) signaling, and perform NAS ciphering and integrity protection. AMF **244** may also support NAS signaling with the UE **202** over an N3 IWF interface.

The SMF **246** may be responsible for SM (for example, session establishment, tunnel management between UPF **248** and AN **208**); UE IP address allocation and management (including optional authorization); selection and control of UP function; configuring traffic steering at UPF **248** to route traffic to proper destination; termination of interfaces toward policy control functions; controlling part of policy enforcement, charging, and QoS; lawful intercept (for SM events and interface to LI system); termination of SM parts of NAS messages; downlink data notification; initiating AN specific SM information, sent via AMF **244** over N2 to AN **208**; and determining SSC mode of a session. SM may refer to the management of a PDU session, and a PDU session or "session" may refer to a PDU connectivity service that provides or enables the exchange of PDUs between the UE **202** and the data network **236**.

The UPF **248** may act as an anchor point for intra-RAT and inter-RAT mobility, an external PDU session point of interconnecting to data network **236**, and a branching point to support multi-homed PDU sessions. The UPF **248** may also perform packet routing and forwarding, perform packet inspection, enforce the user plane part of policy rules, lawfully intercept packets (UP collection), perform traffic usage reporting, perform QoS handling for a user plane (e.g., packet filtering, gating, UL/DL rate enforcement), perform uplink traffic verification (e.g., SDF-to-QoS flow mapping), transport level packet marking in the uplink and downlink, and perform downlink packet buffering and downlink data notification triggering. UPF **248** may include an uplink classifier to support routing traffic flows to a data network.

The NSSF **250** may select a set of network slice instances serving the UE **202**. The NSSF **250** may also determine allowed NSSAI and the mapping to the subscribed S-NSSAIs if needed. The NSSF **250** may also determine the AMF set to be used to serve the UE **202**, or a list of candidate AMFs based on a suitable configuration and possibly by querying the NRF **254**. The selection of a set of network slice instances for the UE **202** may be triggered by the AMF **244** with which the UE **202** is registered by interacting with the NSSF **250**, which may lead to a change of AMF. The NSSF **250** may interact with the AMF **244** via an N22 reference point; and may communicate with another NSSF in a visited network via an N31 reference point (not shown). Additionally, the NSSF **250** may exhibit an Nnssf service-based interface.

The NEF **252** may securely expose services and capabilities provided by 3GPP network functions for the third party, internal exposure/re-exposure, AFs (e.g., AF **260**), edge computing or fog computing systems, etc. In such embodiments, the NEF **252** may authenticate, authorize, or throttle the AFs. NEF **252** may also translate information exchanged with the AF **260** and information exchanged with internal network functions. For example, the NEF **252** may translate between an AF-Service-Identifier and an internal 5GC information. NEF **252** may also receive information from other NFs based on the exposed capabilities of other NFs. This information may be stored at the NEF **252** as structured data, or a data storage NF using standardized interfaces. The stored information can then be re-exposed by the NEF **252** to other NFs and AFs, or used for other purposes such as analytics. Additionally, the NEF **252** may exhibit a Nnef service-based interface.

The NRF **254** may support service discovery functions, receive NF discovery requests from NF instances, and provide the information of the discovered NF instances to the NF instances. NRF **254** also maintains information on available NT instances and their supported services. As used herein, the terms "instantiate," "instantiation," and the like may refer to the creation of an instance, and an "instance" may refer to a concrete occurrence of an object, which may occur, for example, during the execution of program code. Additionally, the NRF **254** may exhibit the Nnrf service-based interface.

The PCF **256** may provide policy rules to control plane functions to enforce them, and may also support a unified policy framework to govern network behavior. The PCF **256** may also implement a front end to access subscription information relevant for policy decisions in a UDR of the UDM **258**. In addition to communicating with functions over reference points as shown, the PCF **256** exhibits an Npcf service-based interface.

The UDM **258** may handle subscription-related information to support the network entities' handling of communication sessions and may store the subscription data of UE **202**. For example, subscription data may be communicated via an N8 reference point between the UDM **258** and the AMF **244**. The UDM **258** may include two parts, an application front end, and a UDR. The UDR may store subscription data and policy data for the UDM **258** and the PCF **256**, and/or structured data for exposure and application data (including PFDs for application detection, application request information for multiple UEs **202**) for the NEF **252**. The Nudr service-based interface may be exhibited by the UDR **221** to allow the UDM **258**, PCF **256**, and NEF **252** to access a particular set of the stored data, as well as to read, update (e.g., add, modify), delete, and subscribe to the notification of relevant data changes in the UDR. The UDM may include a UDM-FE, which is in charge of processing credentials, location management, subscription management, and so on. Several different front ends may serve the same user in different transactions. The UDM-FE accesses subscription information stored in the UDR and performs authentication credential processing, user identification handling, access authorization, registration/mobility management, and subscription management. In addition to communicating with other NFs over reference points as shown, the UDM **258** may exhibit the Nudm service-based interface.

The AF **260** may provide application influence on traffic routing, provide access to NEF, and interact with the policy framework for policy control.

In some embodiments, the 5GC **240** may enable edge computing by selecting operator/3rd party services to be geographically close to a point that the UE **202** is attached to the network. This may reduce latency and load on the network. To provide edge-computing implementations, the 5GC **240** may select a UPF **248** close to the UE **202** and execute traffic steering from the INF **248** to data network **236** via the N6 interface. This may be based on the UE subscription data, UE location, and information provided by the AF **260**. In this way, the AF **260** may influence UPF (re)selection and traffic routing. Based on operator deployment, when AF **260** is considered to be a trusted entity, the network operator may permit AF **260** to interact directly with relevant NFs. Additionally, the AF **260** may exhibit a Naf service-based interface.

The data network **236** may represent various network operator services, Internet access, or third-party services that may be provided by one or more servers including, for example, application/content server **238**.

FIG. **3** schematically illustrates a wireless network **300** in accordance with various embodiments. The wireless network **300** may include a UE **302** in wireless communication with AN **304**. The UE **302** and AN **304** may be similar to, and substantially interchangeable with, like-named components described elsewhere herein.

The UE **302** may be communicatively coupled with the AN **304** via connection **306**. The connection **306** is illustrated as an air interface to enable communicative coupling and can be consistent with cellular communications protocols such as an LTE protocol or a 5G NR protocol operating at mmWave or sub-6 GHz frequencies.

The UE **302** may include a host platform **308** coupled with a modem platform **310**. The host platform **308** may include application processing circuitry **312**, which may be coupled with protocol processing circuitry **314** of the modem platform **310**. The application processing circuitry **312** may run various applications for the UE **302** that source/sink application data. The application processing circuitry **312** may further implement one or more layer operations to transmit/receive application data to/from a data network. These layer operations may include transport (for example UDP) and Internet (for example, IP) operations

The protocol processing circuitry **314** may implement one or more layer operations to facilitate transmission or reception of data over the connection **306**. The layer operations implemented by the protocol processing circuitry **314** may include, for example, MAC, RLC, PDCP, RRC, and NAS operations.

The modem platform **310** may further include digital baseband circuitry **316** that may implement one or more layer operations that are "below" layer operations performed by the protocol processing circuitry **314** in a network protocol stack. These operations may include, for example, PHY operations including one or more hybrid automatic repeat request acknowledgment (HARQ-ACK) functions, scrambling/descrambling, encoding/decoding, layer mapping/de-mapping, modulation symbol mapping, received symbol/bit metric determination, multi-antenna port preceding/decoding, which may include one or more of space-time, space-frequency or spatial coding, reference signal generation/detection, preamble sequence generation and/or decoding, synchronization sequence generation/detection, control channel signal blind decoding, and other related functions.

The modem platform **310** may further include transmit circuitry **318**, receive circuitry **320**, RF circuitry **322**, and RF front end (RFFE) **324**, which may include or connect to one or more antenna panels **326**. Briefly, the transmit circuitry **318** may include a digital-to-analog converter, mixer, intermediate frequency (IF) components, etc.; the receive circuitry **320** may include an analog-to-digital converter, mixer, IF components, etc.; the RF circuitry **322** may include a low-noise amplifier, a power amplifier, power tracking components, etc.; RFFE **324** may include filters (for example, surface/bulk acoustic wave filters), switches, antenna tuners, beamforming components (for example, phase-array antenna components), etc. The selection and arrangement of the components of the transmit circuitry **318**, receive circuitry **320**, RF circuitry **322**, RFFE **324**, and antenna panels **326** (referred generically as "transmit/receive components") may be specific to details of a specific implementation such as, for example, whether the communication is TDM or FDM, in mmWave or sub-6 GHz frequencies, etc. In some embodiments, the transmit/receive components may be arranged in multiple parallel transmit/receive chains, may be disposed of in the same or different chips/modules, etc.

In some embodiments, the protocol processing circuitry **314** may include one or more instances of control circuitry (not shown) to provide control functions for the transmit/receive components.

A UE reception may be established by and via the antenna panels **376**, RFFE **324**, RF circuitry **322**, receive circuitry **320**, digital baseband circuitry **316**, and protocol processing circuitry **314**. In some embodiments, the antenna panels **326** may receive a transmission from the AN **304** by receive-beamforming signals received by a plurality of antennas/antenna elements of the one or more antenna panels **326**.

A transmission may be established by and via the protocol processing circuitry **314**, digital baseband circuitry **316**, transmit circuitry **318**, RF circuitry **322**, RFFE **324**, and antenna panels **326**. In some embodiments, the transmit components of the UE **302** may apply a spatial filter to the data to be transmitted to form a transmit beam emitted by the antenna elements of the antenna panels **326**.

Similar to the UE **302**, the AN **304** may include a host platform **328** coupled with a modem platform **330**. The host platform **328** may include application processing circuitry **332** coupled with protocol processing circuitry **334** of the modem platform **330**. The modem platform may further include digital baseband circuitry **336**, transmit circuitry **338**, receive circuitry **340**, RF circuitry **342**, RFFE circuitry **344**, and antenna panels **346**. The components of the AN **304** may be similar to and substantially interchangeable with like-named components of the UE **302**. In addition to performing data transmission/reception as described above, the components of the AN **304** may perform various logical functions that include, for example, RNC functions such as radio bearer management, uplink and downlink dynamic radio resource management, and data packet scheduling.

FIG. **4** is a block diagram illustrating components, according to some example embodiments, able to read instructions from a machine-readable or computer-readable medium (e.g., a non-transitory machine-readable storage medium) and perform any one or more of the methodologies discussed herein. Specifically, FIG. **4** shows a diagrammatic representation of hardware resources **400** including one or more processors (or processor cores) **410**, one or more memory/storage devices **420**, and one or more communication resources **430**, each of which may be communicatively coupled via a bus **440** or other interface circuitry. For embodiments where node virtualization (e.g., NFV) is utilized, a hypervisor **402** may be executed to provide an execution environment for one or more network slices/sub-slices to utilize the hardware resources **400**.

The processors **410** may include, for example, a processor **412** and a processor **414**. The processors **410** may be, for example, a central processing unit (CPU), a reduced instruction set computing (RISC) processor, a complex instruction set computing (CISC) processor, a graphics processing unit (GPU), a DSP such as a baseband processor, an ASIC, an FPGA, a radio-frequency integrated circuit (RFIC), another processor (including those discussed herein), or any suitable combination thereof.

The memory/storage devices **420** may include a main memory, disk storage, or any suitable combination thereof. The memory/storage devices **420** may include but are not limited to, any type of volatile, non-volatile, or semi-volatile memory such as dynamic random access memory (DRAM), static random access memory (SRAM), erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), Flash memory, solid-state storage, etc.

The communication resources **430** may include interconnection or network interface controllers, components, or other suitable devices to communicate with one or more peripheral devices **404** or one or more databases **406** or other network elements via a network **408**. For example, the communication resources **430** may include wired communication components (e.g., for coupling via USB, Ethernet, etc.), cellular communication components, NEC components, Bluetooth® (or Bluetooth® Low Energy) components, Wi-Fi® components, and other communication components.

Instructions **450** may comprise software, a program, an application, an applet, an app, or other executable code for causing at least any of the processors **410** to perform any one or more of the methodologies discussed herein. The instructions **450** may reside, completely or partially, within at least one of the processors **410** (e.g., within the processor's cache memory), the memory/storage devices **420**, or any suitable combination thereof. Furthermore, any portion of the instructions **450** may be transferred to the hardware resources **400** from any combination of the peripheral devices **404** or the databases **406**. Accordingly, the memory of processors **410**, the memory/storage devices **420**, the peripheral devices **404**, and the databases **406** are examples of computer-readable and machine-readable media.

For one or more embodiments, at least one of the components outlined in one or more of the preceding figures may be configured to perform one or more operations, techniques, processes, and/or methods as outlined in the example sections below. For example, the baseband circuitry as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below. For another example, circuitry associated with a UE, base station, network element, etc. as described above in connection with one or more of the preceding figures may be configured to operate in accordance with one or more of the examples set forth below in the example section.

The term "application" may refer to a complete and deployable package, environment to achieve a certain function in an operational environment. The term "AI/ML application" or the like may be an application that contains some artificial intelligence (AI)/machine learning (ML) models and application-level descriptions. In some embodiments, an AI/ML application may be used for configuring or implementing one or more of the disclosed aspects.

The term "machine learning" or "ML" refers to the use of computer systems implementing algorithms and/or statistical models to perform a specific tasks) without using explicit instructions but instead relying on patterns and inferences. ML algorithms build or estimate mathematical model(s) (referred to as "ML models" or the like) based on sample data (referred to as "training data," "model training information," or the like) to make predictions or decisions without being explicitly programmed to perform such tasks, Generally, an ML algorithm is a computer program that learns from experience with respect to some task and some performance measure, and an ML model may be any object or data structure created after an ML algorithm is trained with one or more training datasets. After training, an ML model may be used to make predictions on new datasets. Although the term "ML algorithm" refers to different concepts than the term "ML model," these terms as discussed herein may be used interchangeably for the present disclosure.

The term "machine learning model," "ML model," or the like may also refer to ML methods and concepts used by an ML-assisted solution. An "ML-assisted solution" is a solution that addresses a specific use case using ML algorithms during operation. ML models include supervised learning (e.g., linear regression, k-nearest neighbor (KNN), decision tree algorithms, support machine vectors, Bayesian algorithm, ensemble algorithms, etc.) unsupervised learning (e.g., K-means clustering, principle component analysis (PCA), etc.), reinforcement learning (e.g., Q-learning, multi-armed bandit learning, deep RL, etc.), neural networks, and the like. Depending on the implementation a specific ML model could have many sub-models as components and the ML model may train all sub-models together. Separately trained ML models can also be chained together in an ML pipeline during inference, An "ML pipeline" is a set of functionalities, functions, or functional entities specific for an ML-assisted solution ML pipeline may include one or several data sources in a data pipeline, a model training pipeline, a model evaluation pipeline, and an actor. The "actor" is an entity that hosts an ML-assisted solution using the output of the ML model inference). The term "ML training host" refers to an entity, such as a network function, that hosts the training of the model. The term "ML inference host" refers to an entity, such as a network function, that hosts the model during inference mode (which includes both the model execution as well as any online learning if applicable). The ML-host informs the actor about the output of the ML algorithm, and the actor decides for an action (an "action" is performed by an actor as a result of the output of an Mt-assisted solution). The term "model inference information" refers to information used as an input to the ML model for determining inference(s); the data used to train an ML model and the data used to determine inferences may overlap, however, "training data" and "inference data" refer to different concepts.

In the future wireless communication systems (e.g., 5G and beyond systems), several verticals can be supported. For example, massive IoT devices, vehicle-to-everything (V2X) communications, wearable devices, private communications, mm-wave (mmW)/terahertz communications, satellite communications, etc. In some of these verticals (e.g., massive IoT and wearables), the end-user devices would be power constrained and have limited computing capability. They may have shorter dwell time, faster network entry requirements due to the compute and power constraints.

In some cellular systems (e.g., 3GPP LTE and 5G systems), network authentication is a centralized process that requires interaction between the end-user device, MME (AMF), and HSS. In some aspects, when a device enters the network, the authentication process over the NAS signaling would happen after the successful RACH (Random Access Channel) Message3, RRC connection request, and the RRC connection complete. This authentication scheme, however, incurs longer latency and substantial network overhead in the network. This can be an issue for lightweight IoT devices that have limited battery life and computation resource for crypto functions. In some of these verticals (massive IoT, wearables), the devices have shorter dwell time, faster network entry requirements due to the compute and power constraints. A simple or lightweight authentication scheme (e.g., as disclosed herein) can be used in connection with the new verticals beyond 5G to have shorter dwell time, faster network entry requirements, and compute/power constraints.

In some embodiments, a hierarchical authentication technique with cryptography-based keys with centralized management can be used along with local distributed schemes. The physical layer (PHY) security authentication based on device fingerprinting using radio frequency (RF) unclonable functions and wireless channel and properties-based techniques are leveraged for authentication in addition to the cryptography-based techniques at the base stations/edge gateway/small cell/RSU. In some aspects, during the network entry process, the full-fledged key exchange (e.g., SIM-based, etc.) is performed. The periodic key refresh and key exchange duty cycles may be relaxed. The base station can be configured to perform periodic PHY layer-based authentication techniques in between key refresh/exchanges. If the base station detects an anomaly or violation, or misbehavior, it may trigger the full-fledged key-based authentication.

In addition, the disclosed techniques introduce multiple security levels for devices. The following three device classes are examples of how a hybrid framework has the flexibility to enable a PHY-based security process (e.g., a device authentication scheme) to be used at varying degrees based on the deployment scenarios and device capability.

Device Class 1: Conventional key-based (e.g., subscriber identity module or SIM-based) authentication mechanism. This class can be used for high-end devices with enough capacity and compute capability.

Device Class 2: Key-based and PHY layer framework authentication. This class can be used for medium-end devices with capacity and latency constraints. For example, V2I and some massive IoT devices

Device Class 3: PHY layer framework authentication. This class can be used for inexpensive massive IOT devices with power, compute, and bandwidth constraints. The access points or gateways can be used for this device class.

Using the disclosed techniques can result in the following technical advantages. Wireless physical layer characteristics are generally harder to manipulate or spoof and hence using these in combination with the crypto-based solution could further enhance the security strength of the overall authentication system. Physical layer-based device authentication techniques between the base station and the end device incur shorter network latency than crypto-based exchanges with MME or AMF. This also eliminates the signaling storm in the core network. PHY layer techniques may also be more lightweight computationally than crypto functions, and hence, may be more suitable for IoT devices. Additionally, physical layer-based device authentication techniques disclosed herein can be done based on the received signals from normal data packets and do not require additional control-plane protocol exchange, or hardware to implement them. The disclosed technique can be implemented on the software based on the measurements available from the lower layers of the received packets.

The disclosed techniques are based on combining the current cryptography-based device authentication framework for 3GPP networks with a new class of device identification techniques that are based on wireless physical properties in a way that reduces overall network latency, signaling overhead, and energy consumption for the devices, while strengthening the overall security.

The disclosed techniques can be based on combining the following two authentication techniques.

Authentication Technique #1 (Cryptography-Based Device Authentication Framework in 3GPP Networks)

In 3GPP-based cellular systems, for example, LTE and 5G, the device authentication can be performed in a centralized framework that is based on symmetric key cryptography. The authentication and key agreement protocol (AKA) can be deployed with both the device (e.g., using a SIM card) and the mobile network operator (e.g., the HSS) demonstrating that they both possess the knowledge of the secret key 'K'. This process is illustrated in FIG. **5**

FIG. **5** illustrates a cryptography-based network authentication framework **500**, according to some example embodiments. Referring to FIG. **5**, the framework **500** can be established in a network that includes UE **502**, base station (e.g., eNB) **504**, MME **506**, and HSS **508**.

At operation **510**, the device (e.g., UE **502**) provides its identifier via a non-access stratum (NAS) message to MME **506**. The identifier could be international mobile subscriber identity (IMSI), global unique temporary identifier (GUTI), or temporary mobile subscriber identity (TMSI). At operation **512**, the MMF **506** passes the identifier and the Serving Network ID to the HSS **508**. These values are then used to generate an authentication vector (AUTN) at the HSS **508**. To compute an AUTN, the HSS **508** can use a random nonce (RAND), the secret key K, and a Sequence Number (SQN) as inputs to a cryptographic function. This function can produce two cryptographic parameters used in the derivation of future cryptographic keys, alongside the expected result (XRES) and an authentication vector (AUTN).

At operation **514**, the authentication vector AUTN (as well as XRES, RAND, and K) is passed back to the MME **506** for storage. In addition, the MME **506** provides (at operation **516**) the AUTN and RAND to the device, which is then passed to the USIM application on the device. The USIM sends AUTN, RAND, the secret key K, and its SQN through the same cryptographic function used by the HSS. The result is labeled as RES, which is sent back to the MME (e.g., at operation **518**). If the XRES value is equal to the RES value, authentication is successful, and the device is granted access to the network.

Authentication Technique #2—Physical Layer (PHY)-Based Techniques for Device Authentication

Using physical layer properties carried in the received wireless signal, a receiver may identify the wireless transmitter with high probability. This is an area of active research and the receiver identification algorithms can be based on Machine Learning. For example, one category of algorithms is called RF fingerprinting or RF-physically unclonable functions (PUF). The general idea is to use the inherent imperfections caused by manufacturing variabilities of the RF components of the wireless devices to uniquely identify wireless devices via an RF signature characteristic of; and unique to, a device. There are many such imperfections in RF components that can be used for this purpose. For example, the local oscillator frequency offset, phase noise, in-phase/quadrature (I-Q) offsets arising from non-idealities in the digital-to-analog converter (DAC), mixer and power amplifier, out of band leakage, etc. can all be used as the unclonable functions to uniquely fingerprint a transmitting device.

In some embodiments, the artificial intelligence (AI) or neural network (NN) classification can be done on the received IQ samples from the device or with some preprocessing (e.g., FFT, etc.). The AI (NN)-based classification can be performed in the following two ways:

(a) Identifying the individual devices from the known set. All the devices are known and pre-registered to the network, and the base station or access point verifies their identity when they try to connect for service.

(b) Classifying a device as an authorized and known device or unknown (and hence, unauthorized) device. If all the authorized devices are known to the network, when a new unknown device is presented, it can be classified as unknown and, hence, unauthorized. This category of techniques may use processing that does not distinguish one authorized device from another.

In some embodiments, the algorithms in the first category (a) that can identify the individual device can be useful in the disclosed authentication framework. The algorithms in the second category (b) may also be sufficient for deployments where the base station does not need to discern the individual device because the access control is the same as long as the device belongs to an authorized group.

In some embodiments, the disclosed techniques are based on a hybrid authentication framework that combines existing cryptography-based authentication, as shown in FIG. **5**, which involves entities in the core network like MME and HSS. The disclosed techniques are also based on a local device authentication scheme using physical layer characteristics of the wireless devices and involve the base station and not the core network. An example communication network that can be used in connection with the disclosed techniques is illustrated in FIG. **6**.

FIG. **6** illustrates an example architecture **600** of a 5G and beyond network, according to some example embodiments. The architecture **600** includes a base station **622** in communication with an authentication network in a core network (e.g., MME **624**). The base station **622** can be configured to communicate with the following types of devices and networks: devices **606** in an industrial IOT network **614**, devices **608** in an enhanced mobile broadband network, devices **610** in a V2X network, and devices **612** in a massive IoT network **620**. Base station **622** can be configured using base station architecture **626** with protocol stacks **628**, **630**, and **632**. The MME **624** can be configured with protocol stack **634**. Network devices **610** and **612** can each be configured with protocol stack **604**, and network devices **606** and **608** can be configured with protocol stack **602**.

In some embodiments, a base station **622** (e.g., access point, eNB, gNB) functions as a local verifier for the compute constrained or power-constrained or short dwell time devices. The physical layer security techniques, such as device fingerprinting using RF unclonable functions, can be used along with cryptography-based techniques to reduce the network overhead and computing burden for computing devices.

The disclosed authentication framework is illustrated in FIG. **8** and can be configured to perform the functionalities illustrated in FIG. **7** and FIG. **8**.

FIG. **7** illustrates a flow chart of functionalities **700** performed in a hierarchical authentication framework, according to some example embodiments. At operation **702**, a device (e.g., UE) is configured for operation in a wireless network. During the network entry process, at operation **704**, a cryptography-based authentication is performed to authenticate the new device. In the meantime, wireless signals from this newly authenticated device are collected by the base station and sent to a new entity in the core network called Physical Layer (PHY) Security Function (PSF). At operation **704**, a timer T**1** for crypto-based authentication is also started.

In some embodiments, PSF can be configured as a network entity responsible to collect training samples (i.e., wireless IQ signal samples) from the cryptographically authenticated devices and perform the necessary training algorithm for the known and authorized devices. The learned model (such as the model for DNN) can be shared with one or more base stations so that the base stations can later use the model to perform a PHY-based security process to physically authenticate the devices. For example, the trained model can be used (e.g., by a base station) to determine an RF signature of a device (e.g., a UE) and authenticate the device based on the determined RF signature. The PSF can also be configured to generate the RF signature of the device based on the signal samples and share such signature with other network entities which can use the device RF signature for device authentication based on the disclosed techniques.

In some embodiments, the periodic key refresh duty cycles are reduced if physical layer-based device authentication schemes are available.

At operation **706**, a timer T**2** for PHY-based security processes (e.g., authentication) is started. In some aspects, the maximum time associated with timer T**1** is greater than the maximum time associated with timer T**2**.

At operation **708**, timer T**2** expires and PHY-based security processes (e.g., authentication) are performed. At operation **710**, a determination is made on whether a device misbehavior is detected based on the authentication techniques. If misbehavior is detected, processing resumes at operation **704**. If no misbehavior is detected, processing continues at operation **712**. At operation **712**, a determination is made on whether timer T**1** has expired. If T**1** has expired, processing resumes at operation **704**. If T**1** has not expired, processing resumes at operation **706**.

In some embodiments, the base station can be configured to perform periodic PHY layer-based authentication techniques in between key refresh to reduce the latency and signaling overhead. To perform the classification at the base station, no additional signaling is used. Thus, this scheme does not cause any network overhead or compute burden on the device side. Instead, the base station periodically runs the classification algorithm based on the learned model from the PSF to verify that the transmitter is indeed the same device it claims previously.

If the base station detects an anomaly or violation (e.g., misbehavior) from a device (e.g., if the physical layer authentication fails for that device), it will trigger the full-fledged crypto-based authentication.

FIG. **8** illustrates a swimlane diagram of example communications flow **800** in a hierarchical authentication framework, according to some example embodiments. Referring to FIG. **8**, the communication flow **800** takes place between the following network entities: UE **802**, base station (e.g., eNB) **804**, MME **806**, HSS **808**, and PSF **810**.

Initially, authentication procedure **812** can be performed. The authentication procedure **812** can include the following functionalities (also discussed in connection with FIG. **5**). At operation **814**, UE **802** communicates device identification information to MME **806**. For example, UE **802** provides its identifier via a non-access stratum (NAS) message to MME **806**. The identifier could be international mobile subscriber identity (IMSI), global unique temporary identifier (GUTI), or temporary mobile subscriber identity (TMSI). At operation **816**, the MMF **806** passes the identifier and the Serving Network ID to the HSS **808**. These values are then used to generate an authentication vector (AWN) at the HSS **808**. To compute an AUTN, the HSS can use a random nonce (RAND), the secret key K, and a Sequence Number (SQN) as inputs to a cryptographic function. This function can produce two cryptographic parameters used in the derivation of future cryptographic keys, alongside the expected result (XRES) and an authentication token (AUTN).

At operation **818**, this authentication vector is passed back to the MME **806** for storage. In addition, the MME **806** provides (at operation **820**) the AUTN and RAND to UE **802**, which is then passed to the USIM application on the device. The USIM sends AUTN, RAND, the secret key K, and its SQN through the same cryptographic function used by the HSS. The result is labeled as RES, which is sent back to the MME **806** (e.g., at operation **822**). If the XRES value is equal to the RES value, authentication is successful, and the device is granted access to the network.

At operation **824**, an indication of a successful crypto-based authentication (e.g., authentication procedure **812**) is communicated from the MME **806** to the PSE **810**. At operation **826**, the PSF **810** communicates a request to the base station **804**. The request is for the base station to start the PHY-based security process. The request can further include a request for the base station to start the collection of signal samples from the UE (e.g., IQ samples) for training. At operation **828**, base station **804** communicates confirmation of the PHY-based security process to the PSF **810**. The base station can also communicate the requested signal samples (e.g., IQ data) to the PSF **810**.

At operation **842**, PSE **810** generates an RF signature of the UE **802** based on the received training samples. At operation **843**, PSF **810** communicates the RF signature to other network nodes (e.g., base station **804**) for subsequent use in a PHY-based security process. Alternatively, PSF **810** trains a machine learning model based on the received signal samples. For example, the machine learning model is trained to associate a specific device (UE) with the corresponding signal samples received from the device. The machine learning model can be trained to associate multiple devices with corresponding signal samples received from such devices. In some embodiments, PSF **810** generates the RF signature based on the received signal samples and trains a machine learning model to associate the device to the determined RF signature. In this regard, after the machine learning model is shared with other network nodes (e.g., base station **804**), the other network nodes can perform a PHY-based security process using the shared machine learning model (e.g., signal samples can be used as input to the model, and the model can indicate the device the signal samples correspond or can indicate whether the device whose samples are entered as input is the correct/authenticated device).

At operations **832**, **834**, **836**, and **838**, periodic PHY-based security (e.g., authentication-related) processes are performed based on the RF signature. The PHY-based security processes can be configured and performed with a periodicity of T**2** **840**.

In some embodiments, the crypto-based authentication procedure **812** can be configured and performed periodically with a periodicity of T**1** **830**. For example, after timer T**1** expires, a subsequent crypto-based authentication procedure (including functionalities **844**, **846**, **848**, **850**, and **852**) is performed. The subsequent crypto-based authentication procedure can be followed by one or more PHY-based security processes **854** (also referred to as a PHY-based authentication procedure).

In some embodiments, the following device security levels can be configured for computing devices using the disclosed techniques (e.g., UE **802**). For example, the following three device classes can be used, where the device classes are associated with varying degrees of usage of the PHY-based authentication procedure during a hybrid scheme.

Device Class 1: Devices using crypto-based (e.g., SIM-based) authentication only. This class can be used for high-end devices with capacity and compute capability that is sufficient to perform the crypto-based authentication procedure. This class may use crypto-based authentication only, it may also elect to use the new hybrid authentication to reap the benefits of low latency and less signaling overhead for the core network, even if it has no computation or energy issue in carrying out the cryptography operation.

Device Class 2: Devices to use crypto-based and PHY-based authentication procedures. This class can be used for medium-end devices with capacity and latency constraints (e.g., V2I devices and some massive IoT devices). These devices can be configured to perform the hybrid authentication scheme to save battery.

Device Class 3: Devices configured primarily with only a PHY-based authentication procedure. This class can be used for inexpensive massive IOT devices with power, compute, and bandwidth constraints. The crypto-based authentication may only be performed at the very first time when the device is newly introduced to the network, and a PHY-based authentication procedure can be the primary way to continue authentication afterward. The base station performs the PHY-based device authentication, and no additional signaling or computation needs to be done by the device.

Wireless Channel Properties-Based Anomaly Detection Schemes

In some embodiments, anomaly detection can be used to detect when unusual or unexpected processes are performed by the device. For example, if the PHY-based device authentication fails, that process can constitute an anomaly. In some aspects, a potential anomaly detection algorithm may monitor the mobility state of the device, or location of the device. This processing can be useful for IoT devices that are placed in a fixed location for a long time to provide service in the field. For example, air quality monitors are placed throughout the city in various locations, which are expected to stay where they are placed indefinitely until they are being serviced. In this regard, location or mobility monitoring to detect when it is moved is a useful technique for anomaly detection.

While any lightweight location or mobility monitoring and tracking algorithms may be used to achieve that, one class of algorithms that are based on the wireless channel properties may be especially useful as it does not require additional sensor hardware. For example, Doppler frequency from the received signal can be used to detect relative motion. The disclosed techniques may not depend or be limited to any class of anomaly detection algorithms.

In some embodiments, if an anomaly is detected based on PHY-based authentication procedures, such detection may activate a crypto-based authentication scheme to make sure the devices are indeed legitimate/authorized devices. In this regard, lightweight PHY-based authentication procedures may be used as the first level of defense while the crypto-based scheme can be used as the second level of defense. In some aspects, when combined, the hybrid framework using crypto-based and PHY-based authentication procedures offers improved processing efficiency, lower latency, and less traffic congestion in the core network.

FIG. **9** illustrates a block diagram of a communication device such as an evolved Node-B (eNB), a new generation Node-B (gNB) (or another RAN node), an access point (AP), a wireless station (STA), a mobile station (MS), or a user equipment (UE), in accordance with some aspects and to perform one or more of the techniques disclosed herein. In alternative aspects, the communication device **900** may operate as a standalone device or may be connected (e.g., networked) to other communication devices.

Circuitry (e.g., processing circuitry) is a collection of circuits implemented in tangible entities of the device **900** that include hardware (e.g., simple circuits, gates, logic, etc.). Circuitry membership may be flexible over time. Circuitries include members that may, alone or in combination, perform specified operations when operating. In an example, the hardware of the circuitry may be immutably designed to carry out a specific operation (e.g., hardwired). In an example, the hardware of the circuitry may include variably connected physical components (e.g., execution units, transistors, simple circuits, etc.) including a machine-readable medium physically modified (e.g., magnetically, electrically, moveable placement of invariant massed particles, etc.) to encode instructions of the specific operation.

In connecting the physical components, the underlying electrical properties of a hardware constituent are changed, for example, from an insulator to a conductor or vice versa. The instructions enable embedded hardware (e.g., the execution units or a loading mechanism) to create members of the circuitry in hardware via the variable connections to carry out portions of the specific operation when in operation. Accordingly, in an example, the machine-readable medium elements are part of the circuitry or are communicatively coupled to the other components of the circuitry when the device is operating. In an example, any of the physical components may be used in more than one member of more than one circuitry. For example, under operation, execution units may be used in a first circuit of a first circuitry at one point in time and reused by a second circuit in the first circuitry, or by a third circuit in a second circuitry at a different time. Additional examples of these components with respect to the device **900** follow.

In some aspects, the device **900** may operate as a standalone device or may be connected (e.g., networked) to other devices. In a networked deployment, the communication device **900** may operate in the capacity of a server communication device, a client communication device, or both in server-client network environments. In an example, the communication device **900** may act as a peer communication device in a peer-to-peer (P2P) (or other distributed) network environment. The communication device **900** may be a eNB, PC, a tablet an STB, a PDA, a mobile telephone, a smartphone, a web appliance, a network router, switch or bridge, or any communication device capable of executing instructions (sequential or otherwise) that specify actions to be taken by that communication device. Further, while only a single communication device is illustrated, the term "communication device" shall also be taken to include any collection of communication devices that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein, such as cloud computing, software as a service (SaaS), and other computer cluster configurations.

Examples, as described herein, may include, or may operate on, logic or a number of components, modules, or mechanisms. Modules are tangible entities (e.g., hardware) capable of performing specified operations and may be configured or arranged in a certain manner. In an example, circuits may be arranged (e.g., internally or with respect to external entities such as other circuits) in a specified manner as a module. In an example, the whole or part of one or more computer systems (e.g., a standalone, client, or server computer system) or one or more hardware processors may be configured by firmware or software (e.g., instructions, an application portion, or an application) as a module that operates to perform specified operations. In an example, the software may reside on a communication device-readable medium. In an example, the software, when executed by the underlying hardware of the module, causes the hardware to perform the specified operations.

Accordingly, the term "module" is understood to encompass a tangible entity, be that an entity that is physically constructed, specifically configured (e.g., hardwired), or temporarily (e.g., transitorily) configured (e.g., programmed) to operate in a specified manner or to perform part or all of any operation described herein. Considering examples in which modules are temporarily configured, each of the modules need not be instantiated at any one moment in time. For example, where the modules comprise a general-purpose hardware processor configured using the software, the general-purpose hardware processor may be configured as respective different modules at different times. The software may accordingly configure a hardware processor, for example, to constitute a particular module at one instance of time and to constitute a different module at a different instance of time.

The communication device (e.g., UE) **900** may include a hardware processor **902** (e.g., a central processing unit (CPU), a graphics processing unit (GPU), a hardware processor core, or any combination thereof), a main memory **904**, a static memory **906**, and a storage device **907** (e.g., hard drive, tape drive, flash storage, or other block or storage devices), some or all of which may communicate with each other via an interlink (e.g., bus) **908**.

The communication device **900** may further include a display device **910**, an alphanumeric input device **912** (e.g., a keyboard), and a user interface (UI) navigation device **914** (e.g., a mouse). In an example, the display device **910**, input device **912**, and UI navigation device **914** may be a touch-screen display. The communication device **900** may additionally include a signal generation device **918** (e.g., a speaker), a network interface device **920**, and one or more sensors **921**, such as a global positioning system (GPS) sensor, compass, accelerometer, or another sensor. The communication device **900** may include an output controller **928**, such as a serial (e.g., universal serial bus (USB), parallel, or other wired or wireless (e.g., infrared (IR), near field communication (NEC), etc.) connection to communicate or control one or more peripheral devices e.g., a printer, card reader, etc.).

The storage device **907** may include a communication device-readable medium **922**, on which is stored one or more sets of data structures or instructions **924** (e.g., software) embodying or utilized by any one or more of the techniques or functions described herein. In some aspects, registers of the processor **902**, the main memory **904**, the static memory **906**, and/or the storage device **907** may be, or include (completely or at least partially), the device-readable medium **922**, on which is stored the one or more sets of data structures or instructions **924**, embodying or utilized by any one or more of the techniques or functions described herein. In an example, one or any combination of the hardware processor **902**, the main memory **904**, the static memory **906**, or the mass storage **916** may constitute the device-readable medium **922**.

As used herein, the term "device-readable medium" is interchangeable with "computer-readable medium" or "machine-readable medium". While the communication device-readable medium **922** is illustrated as a single medium, the term "communication device-readable medium" may include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) configured to store the one or more instructions **924**. The term "communication device-readable medium" is inclusive of the terms "machine-readable medium" or "computer-readable medium", and may include any medium that is capable of storing, encoding, or carrying instructions (e.g., instructions **924**) for execution by the communication device **900** and that causes the communication device **900** to perform any one or more of the techniques of the present disclosure, or that is capable of storing, encoding or carrying data structures used by or associated with such instructions. Non-limiting communication device-readable medium examples may include solid-state memories and optical and magnetic media. Specific examples of communication device-readable media may include non-volatile memory, such as semiconductor memory devices (e.g., Electrically Programmable Read-Only Memory (EPROM), Electrically Erasable Programmable Read-Only Memory (EEPROM)) and flash memory devices; magnetic disks, such as internal hard disks and removable disks; magneto-optical disks; Random Access Memory (RAM); and CD-ROM and DVD-ROM disks. In some examples, communication device-readable media may include non-transitory communication device-readable media. In some examples, communication device-readable media may include communication device-readable media that is not a transitory propagating signal.

Instructions **924** may further be transmitted or received over a communications network **926** using a transmission medium via the network interface device **920** utilizing any one of a number of transfer protocols. In an example, the network interface device **920** may include one or more physical jacks (e.g., Ethernet, coaxial, or phone jacks) or one or more antennas to connect to the communications network **926**. In an example, the network interface device **920** may include a plurality of antennas to wirelessly communicate using at least one of single-input-multiple-output (SIMO), MIMO, or multiple-input-single-output (MISO) techniques. In some examples, the network interface device **920** may wirelessly communicate using Multiple User MIMO techniques.

The term "transmission medium" shall be taken to include any intangible medium that is capable of storing, encoding, or carrying instructions for execution by the communication device **900**, and includes digital or analog communications signals or another intangible medium to facilitate communication of such software In this regard, a transmission medium in the context of this disclosure is a device-readable medium.

The following are some additional example aspects associated with the disclosed techniques.

Example 1 is an apparatus for a physical layer (PHY) security function (PSF) configured for operation in a wireless network, the apparatus comprising: processing circuitry, wherein to configure the PSF for user equipment (UE) authentication in the wireless network, the processing circuitry is to: encode configuration signaling for transmission to a base station, the transmission based on receiving an indication the UE is authenticated via a first authentication process, and the configuration signaling including a request for collection of a plurality of UE signal samples; decode a response message from the base station, the response message including the plurality of UE signal samples collected by the base station; train a machine learning model based on the plurality of UE signal samples, the trained machine learning model associating the authenticated UE with a radio frequency (RF) signature of the UE, the RF signature based on the plurality of UE signal samples; and encode the trained machine learning model for transmission to the base station in connection with a second authentication process of the UE, the second authentication process to authenticate the UE based on the RF signature obtained using the trained machine learning model; and a memory coupled to the processing circuitry and configured to store the plurality of UE signal samples.

In Example 2, the subject matter of Example 1 includes subject matter where the configuration signaling further includes a second request for the base station to initiate the second authentication process of the UE.

In Example 3, the subject matter of Example 2 includes subject matter where the response message further includes a confirmation the second authentication process of the UE is initiated by the base station.

In Example 4, the subject matter of Examples 1-3 includes subject matter where the plurality of UE signal samples include in-phase/quadrature (I/Q) signal samples, and wherein the RF signature is determined based on the I/Q signal samples.

In Example 5, the subject matter of Examples 1-4 includes subject matter where the first authentication process is a cryptographically-based authentication process using an authentication vector, the authentication vector based on a device identifier of the UE, and a serving network identifier of a serving cell of the base station.

In Example 6, the subject matter of Example 5 includes subject matter where the first authentication process is performed with a first periodicity, wherein the second authentication process is performed with a second periodicity, and wherein the first periodicity is higher than the second periodicity.

Example 7 is an apparatus for a base station configured for operation in a wireless network, the apparatus comprising: processing circuitry, wherein to configure the base station for user equipment (UE) authentication in the wireless network, the processing circuitry is to: decode configuration signaling received from a physical layer (PHY) security function (PSF) of the wireless network, the configuration signaling including a request for collection of a plurality of signal samples from a UE, the UE authenticated based on successful completion of a first authentication process;

encode a response message for transmission to the PSF, the response message including the plurality of UE signal samples collected from the UE; decode a trained machine learning model received from the PSF, the trained machine learning model associating the authenticated UE with a radio frequency (RF) signature of the UE, the RF signature based on the plurality of signal samples; and perform a second authentication process of the UE based on the trained machine learning model; and a memory coupled to the processing circuitry and configured to store the configuration signaling.

In Example 8, the subject matter of Example 7 includes subject matter where the first authentication process is a cryptographically-based authentication process using an authentication vector, the authentication vector based on a device identifier of the UE, and a serving network identifier of a serving cell of the base station, and wherein the processing circuitry is to: perform the second authentication process periodically, based on a first periodicity.

In Example 9, the subject matter of Example 8 includes subject matter where the first authentication process is performed periodically, based on a second periodicity, and wherein the first periodicity is different from the second periodicity.

In Example 10, the subject matter of Example 9 includes subject matter where the processing circuitry is to: decode UE capability information received from the UE, the UE capability information including an indication of a device class; and adjust the first periodicity and the second periodicity based on the device class.

In Example 11, the subject matter of Example 10 includes subject matter where the device class indicates the UE is a massive Internet-of-Things (IoT) device, and wherein the processing circuitry is to: suspend subsequent execution of the first authentication process.

In Example 12, the subject matter of Examples 7-11 includes subject matter where the processing circuitry is to: decode a transmission signal received from the UE; and detect a change in a mobility state of the UE based on the transmission signal.

In Example 13, the subject matter of Example 12 includes subject matter where the processing circuitry is to: suspend execution of the second authentication process based on the detected change in the mobility state; and cause execution of the first authentication process.

In Example 14, the subject matter of Examples 7-13 includes subject matter where the processing circuitry is to: detect a failure in the second authentication process; suspend subsequent executions of the second authentication process based on the detected failure; and cause execution of the first authentication process.

Example 15 is a non-transitory computer-readable storage medium that stores instructions for execution by one or more processors of a base station in a wireless network, the instructions to configure the base station for user equipment (UE) authentication in the wireless network and to cause the base station to perform operations comprising: decoding configuration signaling received from a physical layer (PHY) security function (PSF) of the wireless network, the configuration signaling including a request for collection of a plurality of signal samples from a UE, the UE authenticated based on successful completion of a first authentication process; encoding a response message for transmission to the PSF, the response message including the plurality of UE signal samples collected from the UE; decoding a trained machine learning model received from the PSF, the trained machine learning model associating the authenticated UE with a radio frequency (RF) signature of the UE, the RF signature based on the plurality of signal samples; and performing a second authentication process of the UE based on the trained machine learning model.

In Example 16, the subject matter of Example 15 includes, the operations further comprising: performing the second authentication process periodically, based on a first periodicity.

In Example 17, the subject matter of Example 16 includes subject matter where the first authentication process is performed periodically, based on a second periodicity, and wherein the first periodicity is different from the second periodicity.

In Example 18, the subject matter of Example 17 includes, the operations further comprising: decoding UE capability information received from the UE, the UE capability information including an indication of a device class; and adjusting the first periodicity and the second periodicity based on the device class.

In Example 19, the subject matter of Example 18 includes subject matter where the device class indicates the UE is a massive Internet-of-Things (IoT) device, and the operations further comprising: suspending subsequent execution of the first authentication process.

In Example 20, the subject matter of Examples 15-19 includes, the operations further comprising: decoding a transmission signal received from the UE; detecting a change in a mobility state of the UE based on the transmission signal; suspending execution of the second authentication process based on the detected change in the mobility state; and causing execution of the first authentication process.

Example 21 is at least one machine-readable medium including instructions that, when executed by processing circuitry, cause the processing circuitry to perform operations to implement any of Examples 1-20.

Example 22 is an apparatus comprising means to implement any of Examples 1-20.

Example 23 is a system to implement any of Examples 1-20.

Example 24 is a method to implement any of Examples 1-20.

Although an aspect has been described with reference to specific exemplary aspects, it will be evident that various modifications and changes may be made to these aspects without departing from the broader scope of the present disclosure. Accordingly, the specification and drawings are to be regarded in an illustrative rather than a restrictive sense. This Detailed Description, therefore, is not to be taken in a limiting sense, and the scope of various aspects is defined only by the appended claims, along with the full range of equivalents to which such claims are entitled.

What is claimed is:

1. An apparatus for a base station configured for operation in a wireless network, the apparatus comprising:

processing circuitry, wherein to configure the base station for user equipment (UE) authentication in the wireless network, the processing circuitry is to:

decode configuration signaling received from a physical layer (PHY) security function (PSF) of the wireless network, the configuration signaling including a request for collection of a plurality of signal samples from a UE, the UE authenticated based on successful completion of a first authentication process;

encode a response message for transmission to the PSF, the response message including the plurality of UE signal samples collected from the UE;

decode a trained machine learning model received from the PSF, the trained machine learning model associating the authenticated UE with a radio frequency (RF) signature of the UE, the RF signature based on the plurality of signal samples; and

perform a second authentication process of the UE based on the trained machine learning model; and

a memory coupled to the processing circuitry and configured to store the configuration signaling.

2. The apparatus of claim **1**, wherein the first authentication process is a cryptographically-based authentication process using an authentication vector, the authentication vector based on a device identifier of the UE and a serving network identifier of a serving cell of the base station, and wherein the processing circuitry is to:

perform the second authentication process periodically, based on a first periodicity.

3. The apparatus of claim **2**, wherein the first authentication process is performed periodically, based on a second periodicity, and wherein the first periodicity is different from the second periodicity.

4. The apparatus of claim **3**, wherein the processing circuitry is to:

decode UE capability information received from the UE, the UE capability information including an indication of a device class; and

adjust the first periodicity and the second periodicity based on the device class.

5. The apparatus of claim **4**, wherein the device class indicates the UE is a massive Internet-of-Things (IoT) device, and wherein the processing circuitry is to:

suspend subsequent execution of the first authentication process.

6. The apparatus of claim **1**, wherein the processing circuitry is to:

decode a transmission signal received from the UE; and

detect a change in a mobility state of the UE based on the transmission signal.

7. The apparatus of claim **6**, wherein the processing circuitry is to:

suspend execution of the second authentication process based on the detected change in the mobility state; and

cause execution of the first authentication process.

8. The apparatus of claim **1**, wherein the processing circuitry is to:

detect a failure in the second authentication process;

suspend subsequent executions of the second authentication process based on the detected failure; and

cause execution of the first authentication process.

9. A non-transitory computer-readable storage medium that stores instructions for execution by one or more processors of a base station in a wireless network, the instructions to configure the base station for user equipment (UE) authentication in the wireless network and to cause the base station to perform operations comprising:

decoding configuration signaling received from a physical layer (PHY) security function (PSF) of the wireless network, the configuration signaling including a request for collection of a plurality of signal samples from a UE, the UE authenticated based on successful completion of a first authentication process;

encoding a response message for transmission to the PSF, the response message including the plurality of UE signal samples collected from the UE;

decoding a trained machine learning model received from the PSF, the trained machine learning model associating the authenticated UE with a radio frequency (RF) signature of the UE, the RF signature based on the plurality of signal samples; and

performing a second authentication process of the UE based on the trained machine learning model.

10. The non-transitory computer-readable storage medium of claim **9**, the operations further comprising:

performing the second authentication process periodically, based on a first periodicity.

11. The non-transitory computer-readable storage medium of claim **10**, wherein the first authentication process is performed periodically, based on a second periodicity, and wherein the first periodicity is different from the second periodicity.

12. The non-transitory computer-readable storage medium of claim **11**, the operations further comprising:

decoding UE capability information received from the UE, the UE capability information including an indication of a device class; and

adjusting the first periodicity and the second periodicity based on the device class.

13. The non-transitory computer-readable storage medium of claim **12**, wherein the device class indicates the UE is a massive Internet-of-Things (IoT) device, and the operations further comprising:

suspending subsequent execution of the first authentication process.

14. The non-transitory computer-readable storage medium of claim **9**, the operations further comprising:

decoding a transmission signal received from the UE;

detecting a change in a mobility state of the UE based on the transmission signal;

suspending execution of the second authentication process based on the detected change in the mobility state; and

causing execution of the first authentication process.

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